Incentivizing Quality in Hospice Care

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Abstract

I study policies that can improve hospice quality and their implications for Medicare spending. Hospices receive fixed Medicare payments per patient per day. High-quality care is costly, but consistently providing it builds a stronger quality-history for a hospice. I show that better quality-history positively influences consumer choice. I build a structural model of hospices competing by dynamically choosing quality to build up their quality-histories, and I use it to recover the hospice cost function and run the following counterfactuals. Lower Medicare reimbursement reduces quality. Partially tying reimbursement to quality reduces Medicare spending without sacrificing equilibrium quality. Stronger consumer responses to quality-history – which may be achieved through a hospice review website – improve equilibrium quality and amplifies reimbursement's effects on quality.

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1 Introduction

I study market design improvements in the US hospice industry that may improve quality while being cognizant of Medicare spending. This is useful for several reasons. First, the US hospice industry is large and growing – over 1.6 million Medicare patients used hospices in 2022, and Medicare spent around $21 billion on hospice services in 2019. Second, hospice care is a low-cost alternative that can significantly reduce end-of-life spending by moving many patients away from painful ineffective care; however, we need to incentivize good hospice quality for patients to switch.[1] Finally, this will shed light on policy design for "regulated price" healthcare settings, where the government fixes reimbursement rates and providers compete on non-price dimensions such as quality. Outside the US, such regulated-price healthcare markets are particularly common in Europe as well as the rest of the world.

I argue that there are two sets of tools we may use to affect equilibrium quality and payor spending. First, we can use alternative reimbursement terms and designs. Hospice care is a regulated-price industry where hospices are paid a fixed rate per patient-day by Medicare. Thus, one way of incentivizing higher quality while balancing spending may be to adjust reimbursement rates and schemes. Second, in healthcare provider competition, consumers often have only imperfect information on provider quality in the current period. As a result, they often rely on a provider's quality history to choose between different providers, for reasons such as firm goodwill/brand capital and reputation. A provider which has consistently chosen high quality in the past is more likely to be picked by the consumer, and therefore have higher market share. The importance of quality history introduces dynamics into providers' quality choices, as current quality choice affects future quality-history and therefore market share. Such dynamics unlocks information-based policy tools that may be used to incentivize higher quality without higher per-patient spending.

Using a dynamic oligopoly model of the California hospice industry, I show that hospices dynamically choose quality to improve their quality-history and attract consumers. Then, I use my model and estimates to evaluate counterfactual policies. Reducing Medicare's hospice reimbursement rates worsens hospice quality. A hybrid per-day per-visit Medicare reimbursement scheme can lower Medicare spending without worsening equilibrium hospice quality. Strengthening the effect of quality-history on consumer choice – such as having a hospice review website that makes such history more accessible and salient – leads to higher equilibrium quality. Stronger quality-history effect also amplifies the impact of Medicare reimbursement on hospice quality.

Hospice care is given to dying patients at their residences, where hospice staff administer pain medication and help with living arrangements. More visits by the hospice staff can be thought of

[1] Concerns have been raised about hospice quality across the US. The Department of Health and Human Services has released a report citing numerous deficiencies in hospice quality: https://oig.hhs.gov/newsroom/media-materials/media-materials-2019-hospice/

as implying higher quality – it means the hospice is checking up on patients more and adjusting for day-to-day difficulties. Therefore, my measure of hospice quality is mean visits-per-patient-day. There is no price competition because most hospice patients are covered by Medicare, and Medicare pays hospices a fixed per-day rate for each patient.

Furthermore, when it comes to choosing healthcare providers, consumers often have at best imperfect information on current quality. As a result, they may rely on a hospice's quality-history.[2] A hospice with a strong track record of high quality care in the past is more likely to be known and chosen within the community. This suggests that a hospice can build up its quality-history by consistently choosing high quality over time, and use this to attract more patients. The lack of price competition makes quality-history particularly important for consumers. I provide descriptive evidence to confirm my claim about visits-per-patient-day being an appropriate measure of quality and effort, as well as evidence suggesting that quality-history is important for consumer choice of hospices.

To study information-based and reimbursement-based policy tools, I build a structural model of hospice demand by consumers and dynamic quality choice by hospices. For tractability, the quality-history of a hospice is captured as a discounted-sum of its past quality choices. On the demand side, each consumer makes a discrete choice from a set of hospices in her market, taking into account their quality-histories. On the supply side, hospices play a dynamic game in an oligopoly setting where they choose quality every period. They take into account that higher quality will lead to improved quality-history and market share but also higher cost per consumer.

I estimate this model using annual hospice-level data from 39 counties in California for 2002-2018. The estimation proceeds as follows. First, I estimate a nested-logit discrete choice demand model allowing for persistent unobserved demand shocks. This lets me pin down the magnitude and discounting of quality-history for consumer choice. Second, I estimate a dynamic oligopoly model of quality choice following the methods of Bajari et al. (2007) and Pakes et al. (2007). By incorporating additional data on hospice cost, this estimation procedure recovers the hospice cost function. The demand estimates show that quality-history plays a significant role in consumer

[2]Why might a consumer care about quality-history in the presence of imperfect information on current quality? If the consumer had perfect information on current quality, then they know the quality they will receive upon choosing a provider, so past quality does not matter. But if the consumer has no/imperfect information on current quality – as is often the case when choosing providers – then quality-history plays an important role. A hospice which has chosen high quality in the past has built up its firm goodwill, brand capital, and recognition in the community through repeated good service, and so is more likely to be chosen/referred to the patient. A quality-history reflects years of service of the firm, which may also positively influence consumer choice. Consumers can use quality-history to form expectations about current quality, and without having to track all possible state variables that influence equilibrium quality choice.

Information on current quality and quality-history comes to consumers from different sources. A firm's quality-history may be acquired by talking to social workers, physicians, and other members of the community. To get information on current quality, consumers need to talk to/visit a hospice and track down its current patients, etc. This makes information on current quality harder to obtain and interpret, and is also likely to be noisy.

choice. The hospice cost estimates show that the cost of an additional visit ranges from $153 to $256, depending on the hospice and the county it is located in.

Using the estimated demand and cost parameters, I solve the model via policy iteration. I use this to solve for equilibrium quality choices under various counterfactual policies involving reimbursement-based and information-based policy tools. I first focus on reimbursement-based policy tools.

First, I show that higher Medicare rates leads to higher equlibrium quality. The intuition is that higher rates means each patient is now more financially valuable to the hospice, so it is optimal to choose higher quality (and incur higher costs) to attract more patients. Higher rates can also incentivize entry by new hospices, and the resulting competition further improves hospice quality; correspondingly, lower rates can induce exit and worsen quality further. Therefore, CMS should be cognizant of lower reimbursement reducing quality of service. Higher quality from higher reimbursement can also cause patients to switch away from even more expensive care, such as that documented in Gruber et al. (2023), potentially generating net savings for Medicare despite higher per-diem rates.

Second, I study an alternative reimbursement scheme structure for Medicare. I compare the current per-day Medicare reimbursement scheme with a hybrid per-day per-visit reimbursement scheme. The latter is found to achieve the same equilibrium quality level but with lower Medicare spending. The intuition is that while before the payoff from increasing quality was gaining/defending market share, under the hybrid scheme hospices get an additional payoff through the per-visit reimbursement, causing equilibrium quality to be higher. However, I also find that lifetime profits of hospices decline, so it may increase exit incentives. Therefore, partially tying reimbursement to quality can lower spending without harming quality, while taking care to not force hospice exit.

For my next counterfactuals, I consider information-based policy tools. In 2017, CMS launched Hospice Compare (now part of Care Compare) to provide consumers with information on hospice quality-history. However, it is difficult to empirically measure the effect of this website due to the quality-measure rollout happening during COVID, as well as this being a national rollout. This raises two research questions. First, how can increased accessibility and salience of quality-history information affect consumer choice and equilibrium hospice quality? Second, how does such information interact with Medicare reimbursement policies to shape equilibrium quality? Understanding these effects has implications for quality-reporting platforms in healthcare and can inform efforts to further improve Hospice Compare.

I examine what happens when quality-history has a stronger effect on consumer choice. Strengthening the quality-history effect on consumer choice leads to higher equilibrium hospice quality. This is because, for the hospice, the marginal gain from raising quality increases: an increase in

quality leads to a larger increase in future sales, so it is optimal to incur the higher cost from higher quality. There is also reallocation of market share from less-efficient to more-efficient hospices. Total Medicare spending on hospices rises due to a market expansion effect, as the mix of stronger quality-history effect and higher equilibrium quality pulls people out of the outside option into hospice care. This suggests that promoting and incentivizing participation on Hospice Compare can improve equilibrium quality with little additional deployment cost to CMS.

Finally, I study the interaction between quality-history effect and Medicare's reimbursement rates. First, stronger quality-history effect on consumer choice amplifies the impact of Medicare rates on equilibrium quality. Second, we can strengthen quality-history effect and reduce per-diem Medicare rates to maintain the same equilibrium quality, but it leads to an increase in total Medicare spending on hospices. This is due to a market expansion effect, as the stronger quality-history effect draws in consumers from the outside option into hospice care.

For several counterfactuals, I find that the market expansion effect leads to total Medicare hospice spending rising. This does not necessarily mean that total Medicare (or payor) spending rises, because it depends on the outside option patients are moving away from. If the outside option involves Medicare (or other payors) spending on expensive ineffective treatment, then this represents a net decline in spending.

Contributions and related literature: First, I contribute to a sparse literature on the hospice industry and its regulations. I am the first to structurally estimate the supply-side of this industry. Prior work on the hospice industry include Chung and Sorensen (2018), who estimate a nested logit demand model to examine market expansion from hospice entry; Dalton and Bradford (2019), who study length of stay of patients in non-profit and for-profit hospices; and Gruber et al. (2023), who show that hospices save money for Medicare by offsetting alternative expensive care for Alzheimer's and Dementia patients.

Second, I add to the literature of modeling provider choice and competition. In contrast to prior literature, I contribute by allowing quality-history to affect consumer choice alongside current quality, which in turn requires providers to choose quality dynamically. This phenomenon is understudied in the health economics literature and potentially extends to other industries, particularly those with non-contractible quality or where a firm's brand capital plays a key role. Furthermore, I argue that accounting for quality-history in the demand model allows for the correct variation in the data to estimate the role of current and past quality in demand estimation. Prior literature project market share on current quality to estimate demand, but it is unclear what variation in the data is being used to estimate the effect of current quality when quality-history is entirely omitted from the function. Some papers on provider choice and competition include Einav et al. (2018), Hackmann (2019), Hotz and Xiao (2011), Lin (2015), and Xiao (2010). More broadly, my work is connected to the use of information to influence consumer choice and provider qual-

ity in healthcare settings. Dafny and Dranove (2008) show how consumers learn about Medicare HMO quality through both "market-based" learning (from word-of-mouth, providers, personal experience, etc.) and exogenous information provision (through public HMO report cards), and this leads to changes in market share. Jin and Leslie (2003) and Jin and Sorensen (2006) show that consumers switch to higher-quality firms (for restaurants and health plans, respectively) when public report cards are released, while Chartock (2025) shows that consumers switch to higher-quality physicians after quality-ratings disclosure. Sorensen (2006) finds social learning about health plan choices within academic departments, underscoring the importance of learning through word-of-mouth about quality over time.

Third, my work is connected to the literature on advertising and brand capital management. Papers in this area study how repeated advertising flows build up a firm's advertising/goodwill/brand capital stock, and how this stock influences consumer choice of products. I show that brand capital is also built up through repeated quality of service. Besides the conceptual similarity, my choice of functional forms for quality and quality-history is similar to theirs for goodwill stock/brand capital. Some relevant papers in that regard are Abi-Rafeh et al. (2026), Borkovsky et al. (2017), Dubé et al. (2005), and Dubois et al. (2018). See Bronnenberg et al. (2022) for an overview of brand capital and firm incentives.

2 Industry background

Hospices provide palliative care to terminally ill patients, the majority of whom die within 1 week to 3 months in my dataset. Patients who enroll in hospice typically no longer receive curative treatment. Most patients in the dataset are covered by Medicare's reimbursement scheme, which pays hospices a fixed amount for every day the patient is enrolled. In return, hospice staff provide care to the patient by visiting her at her residence. I claim that a hospice can build up its quality-history by consistently providing high quality over time.

2.1 Hospice care provision

Unlike other medical providers, hospices typically provide care at the residence of the patient. Nurses and home health aides make the majority of hospice visits.[3] The hospice staff make multiple visits to a patient throughout the patient's enrollment period. Administrators of a hospice decide the number of visits, and set up the schedule for each nurse. A day for a hospice nurse

[3]Hospices employ several types of staff: registered nurses, licensed vocational nurses, home healthcare aides, physicians, social workers and chaplains. Physicians rarely make visits, and usually deal with reports on the patients from nurses. Social workers help the patient and her family with paperwork and choices.

involves driving to several patients as determined by the administrator, giving them care, writing up reports for the hospice physician and administrator, and refilling supplies.

The care given by hospice nurses is relatively low-skill compared to other medical providers. Hospices are not responsible for curative treatment; they are tasked with pain management and ease-of-living for a dying patient. Examples of hospice care include administering pain medication, providing medical supplies like oxygen and bandages, dressing bedsores, giving physical and speech therapy, helping with bathing and feeding, temporarily substituting as the primary caregiver, tending to emotional and spiritual needs of the patient, and giving grief counselling to the family after the patient passes away.[4]

It is important to note that the hospice decides how many visits to make to each patient and what care is to be provided during each visit. As is explained below, the patient does not incur any additional cost for more visits, and so would plausibly like as many visits as possible.

How a hospice is chosen by a patient and her family vary. Sometimes a social worker at the hospital gives the patient a list of hospices, and suggestions on which to choose based on past experience; the same can be done by the patient's physician. The family can also search for hospices online (aided by online reviews and forums), or can rely on word-of-mouth from other families that have needed hospice services in the past.

2.2 Payers and reimbursement schemes

For hospice reimbursement, I focus my analysis on eligibility rules and reimbursement schemes for Medicare only. While hospice services can be paid for by a variety of payers – Medicare, Medi-Cal (California's Medicaid), private insurance, charity, self-pay, etc. – the overwhelming majority of patients are paid for by Medicare (see Table 1). Furthermore, Medi-Cal's reimbursement is essentially the same as Medicare's.

For a patient covered by Medicare, hospice care is essentially free. There is no out-of-pocket cost for enrolling into or receiving visits from a hospice. Patients have to give a copay of up to $5 for outpatient drugs required for pain management. This means that the "price" of choosing a hospice does not vary across hospices in the patient's choice set, and moreover can be thought of as zero. The patient generally has to turn down curative treatment to enroll into hospice care via Medicare.

For the hospice, enrolling a patient covered by Medicare results in the hospice getting paid a

[4]Hospice staff do not function as the primary caregiver of the patient - someone in the patient's residence (a family member if the patient resides at home, or a nurse if the patient resides in a nursing facility) has to be the primary caregiver. But hospice nurses assist the primary caregiver, and sometimes may substitute in for a period of time to give respite to the primary caregiver.

fixed rate for every day the patient is enrolled (i.e. a per-diem rate).[5] It is important to note that the fixed rate is set exogenously and does not depend on the number of visits that the hospice staff makes. To contrast extreme examples, if a patient enrolls in a hospice for 10 days, the hospice gets paid a fixed amount for each of those 10 days irrespective of how many visits they make.[6]

The Medicare rate is determined as follows. Medicare sets a national per-diem rate that is split into a "labor component" and a "non labor component". Medicare also creates wage indices for each Metropolitan Statistical Area (MSA), such that a MSA with high wages will have a high wage index. The per-diem reimbursement rate of a hospice is the non-labor component and the labor component times the wage index of the hospice's MSA.[7] These rates are updated every year.

Payer	% of patients
Medicare	83.7
Medi-Cal	7.37
Private insurance	6.37
Selfpay	1.65
Charity	0.82

Table 1: Percentage of patients in dataset covered by each payer type.

2.3 Hospice competition and quality-history

The institutional details indicate that hospices primarily compete in quality. Hospices do not compete in prices as they are paid a fixed rate by Medicare. Hospices also do not compete on location conditional on being in the same market – a patient's distance to the hospice is unlikely to play a role in her hospice choice as the staff visit the patient at her residence. Higher quality is more costly to provide, and hospices are not directly reimbursed for higher quality, so the main economic motive for providing higher quality is to attract more patients. The absence of price competition makes quality particularly important in consumers' decisions.

Furthermore, I hypothesize that a hospice's quality-history plays a key role in consumer choice.

[5] Medicare also distinguishes between different types of hospice care. Specifically, it draws a distinction between routine care (which is the care described above), inpatient care (where a patient moves to an inpatient ward of a hospice for intense treatment) and continuous respite care (where the hospice takes over as the primary caregiver). The majority of patients in my sample are given routine care; I show in Appendix E that of all the "days of care" provided by a hospice within each year, over 99% is dedicated to routine care. Medicare allows the per-diem rate to vary substantially based on whether the care provided is routine home care, inpatient care, or continuous care.

[6] There are additional dimensions of the Medicare hospice reimbursement scheme that I do not study in this paper: see Appendix G.4.2.

[7] To work out an example: for FY 2021, the labor component of routine home care is \$136.9, and the non-labor component is \$62.35. Suppose a hospice is in a county with a wage index of 0.8. The reimbursement rate of the hospice is approximately $136.9 \times 0.8 + 62.35 = \$171.87$. See Appendix K.1 for an example of Medicare's reimbursement rates and its variation across counties.

Consumers choosing healthcare providers often have imperfect information about current quality and may therefore rely on a provider's quality-history. With perfect information, consumers would know the quality they would receive, making past quality irrelevant. With limited or no information, however, quality-history can influence their choice. Quality-history can affect consumer choice through several channels. First, a hospice that has consistently provided high-quality care builds goodwill, brand capital, and recognition in the community. These assets make consumers more likely to choose the hospice and others more likely to refer patients to it.[8] Second, quality-history reflects a hospice's years of service, which may independently influence consumer choice. Third, past quality helps consumers form expectations about current care without tracking every state variable that affects a provider's equilibrium quality choice. Such state-variables can include market-level factors like competition trends, and persistent firm-level factors such as cost-efficiency or altruism that induce a provider to provide higher quality.

Information about current quality and past quality can come from different sources. A firm's quality-history may be obtained by talking to social workers, physicians, and other members of the community. (Since a hospice has few repeat interactions with the same family, presence of intermediaries such as social workers is how information about a hospice's past quality choices can persist over time.) Assessing current quality requires contacting or visiting the hospice and its current patients. This information can be difficult to obtain and interpret and is likely to be noisy.

Unsurprisingly, there is anecdotal evidence that quality-history matters for consumer choice. When looking for advice on how to choose a hospice, one common suggestion is to choose hospices that are reputable and have served the community well over a long time.[9]

In Section 3 I provide descriptive evidence suggesting that quality-history is important for hospice choice by consumers.

3 Data

The dataset for this paper comes from firm-level data on hospices in California for 2002-2018. A market is defined to be at the level of a county, and I restrict my estimation to 39 counties.

[8] This has parallels with the intangible brand capital literature - for example Bronnenberg et al. (2022) discuss brand equity and reputation, and how these create incentives for firms to repeatedly supply high quality over time.

[9] Some websites which give suggestions on how hospices are chosen are listed in Appendix K alongside relevant quotes. A representative quote from AmericanHospice.org reads: "What do others say about this hospice? Get references both from people you know and from people in the field – e.g., local hospitals, nursing homes, clinicians. Ask anyone that you have connections to if they have had experience with the hospice and what their impressions are. Geriatric care managers can be a particularly good resource, as they often make referrals to hospices and hear from families about the care that was provided."

Similarly, HospiceFoundation.org states: "Seek professional opinions. Ask clinicians, professional caregivers at nursing homes, geriatric care managers, or end-of-life doulas about their experience with a hospice. Talk to friends, family, and neighbors who have used hospice services and get their opinions about the experience with a provider."

Total firms in a market range from 1 to 23, with moderate entry and little exit. I use these data to construct firm-level measures of quality, defined to be visits-per-patient-day made by a hospice in a year. I provide descriptive evidence showing that visits-per-patient-day is an appropriate measure of quality and suggesting the importance of quality-history for consumer choice.

3.1 Data sources

The main dataset comes from Home Health Agencies And Hospice Annual Utilization Reports compiled by California Department of Health Care Access and Information (HCAI). Hospices in California are required to submit yearly utilization reports to HCAI where they report firm-level statistics for the year. This includes measures such as current location, total number of patients, total visits per year (broken down by type of staff), hospice characteristics, aggregate characteristics of patient pool, annual expenses, etc. I collect and clean utilization reports of hospices for 2002-2018. I complement this with additional data sources on population sizes (by age), mortality rates, and Medicare hospice reimbursement rates for California over the same time period.

Additional details on how I deal with nonresponses, outliers, and firm IDs are specified in Appendix A.

3.2 Market definition

A market is defined to be at the level of a county.[10] Market size is constructed by multiplying the population size of each age for a market with the corresponding mortality rate; that is, market size is the expected number of deaths happening in a market in a year.

I focus on 39 counties for my project. While California has 58 counties, I have to drop the remainder for the following reasons. First, several counties do not see any hospice presence over the course of my dataset. Second, a few counties (6) have a very large number of hospices. Markets with many small sellers have very different dynamics and competition than those with few sellers. For instance, search frictions and consideration choice sets may mean that consumers do not consider all firms in a market, which would make estimating competitive effects of rivals more difficult. It could also lead to very different ways of competing with rival hospices. These markets are also much larger, and so competition may be more localized than a county. Finally, these six counties exhibit such rapid growth in number of hospices that they are likely to be exhibiting nonstationary dynamics, which my stationary dynamic model is not well-equipped to handle. For

[10] While some hospices offer services to multiple counties, I can rule out clusters of counties as a market definition using data. Specifically, the dataset contains a breakdown of a hospice's patient pool by patients' counties of residence. Most of a hospice's patients originate from the county where the hospice is located; see Appendix E.4. The only exception are Yuba and Sutter counties which have a large overlap of patients, and so I combine them into a single market.

instance, between 2002 to 2017, San Bernardino went from 6 to 84 hospices. A separate paper (Alam (2025)) is studying hospice efficiency and entry-exit in all California counties. I discuss these points in more detail in Appendix E.4.

How representative are markets with 23 or fewer hospices of the US hospice industry, as is the case with my sample? For that, I look at the number of hospices by 3-digit zip code in the US for the year 2025.[11] The 90th percentile of that distribution has 17 hospices in a 3-digit zip code.

Such sample selection is in line with many dynamic oligopoly papers such as Collard-Wexler (2013) and Lin (2015). Please see Appendix E.4 for more details and discussions on market selection and a list of the counties I focus on.

3.3 Summary statistics on competition and patients

I now present some summary statistics on competition and patient composition within my sample. The tables below and in Appendix E show distributions of various key variables; the main takeaways are as follows. First, the vast majority of hospices have over 100 patients. This assuages worries that my key measure of quality - visits per patient day - is mostly driven by fluctuating patient severity, since we would expect that variations in patient severity would even out over a large number of patients over the course of the year. Second, the median firm count is 2 with the 75th percentile at 4, suggesting that an oligopoly setting is the correct mode of analysis instead of perfect competition. Third, while the total entry and exit over the 16 years of my data is significant, there is very little turnover in a market within an individual year. In my model, I will treat exit as effectively exogenous due to the lack of variation in my data. Fourth, the vast majority of patients have a length of stay between a week and 3 months; see Appendix E for more details. Finally, Appendix E also does a breakdown of patients by diagnosis, revealing that roughly 40% of the patients are suffering from cancer.

Table 2: Distribution of market-level characteristics.

Variable	Min	10%	25%	50%	75%	90%	Max
Firm count	1.0	1.0	1.0	2.0	4.0	7.0	23.0
Entry count	0.0	0.0	0.0	0.0	0.0	1.0	5.0
Exit count	0.0	0.0	0.0	0.0	0.0	1.0	3.0
Medicare rates	134.48	151.88	178.29	212.41	255.25	284.52	375.92

Notes: Each observation is at the county-year level. Medicare per-day rates are inflation-adjusted.

The dataset contains information on hospice-specific services and characteristics. To give a

[11] I use CMS data from https://data.cms.gov/provider-characteristics/hospitals-and-other-facilities/hospice-enrollments/data

snapshot, in 2017 there were 171 hospices in my dataset; of these, 12 had an inpatient unit, 15 offered special pediatric services, 2 offered adult day care services, and 95 were for-profit. The dataset also breaks down hospices by agency type: 110 hospices were free-standing, 27 also had a home-health unit, and 31 were hospital-based (a few hospices had agency type classified as "Other"). I control for these characteristics in my demand estimation.[12]

3.4 Measure of hospice quality

I discussed in Section 2.3 that hospices compete in quality. I now explain the measure of quality that I will use in my estimation.

I claim that the frequency with which a hospice visits its patients (visits-per-patient-day) reflects the quality and effort the hospice provides. This is based on several arguments.

1. I provide descriptive evidence showing that visits-per-patient-day by a hospice both responds to the level of competition the hospice faces and correlates with the hospice having higher market share (Tables 4 and 5). This suggests visits-per-patient-day is a measure of quality, since hospice quality should rise when competition increases, and hospices choosing higher quality should get higher market share.

2. A hospice which makes more visits is constantly checking and adjusting to the patient's condition, can manage symptoms better, and is likely to arrive quickly in case of an emergency. Such a hospice is also able to provide more respite to the patient's primary caregiver, which may be highly valued by the family.

3. Visits-per-patient-day is plausibly a good proxy for the general effort exerted by a hospice. Visits are the most costly part of care as the hospice has to hire extra nurses and pay the fuel cost of traveling. If a hospice is incurring this cost already, it is unlikely that they are then shirking on within-visit care. Therefore, it is also likely that number of visits is correlated with the general effort of the hospice, and so with other unobserved dimensions of quality.[13]

4. Visits-per-patient-day is in line with several measures of hospice quality tracked by CMS (see Appendix M for details). Among other indicators, CMS tracks: Hospice Visits in Last (three) Days of Life, Gaps in Skilled Nursing Visits, Skilled Nursing Care Minutes per Routine Home Care Day, Skilled Nursing Minutes on Weekends, Training Family To Care For

[12]The dataset also states how many patients of a hospice were discharged to a different hospice. The distribution of this variable as a share of total patients is reported in Table 37. Over 40% of hospices do not discharge any patient to a different hospice, and of the remainder the fraction is mostly less than 2%. In general, there is very little variation.

[13]Hospice care mostly involves relatively low-skill care such as giving pain medication, making adjustments to living arrangements, and helping out the patient with certain physical activities. As such, it is unlikely that there will be much heterogeneity across hospices regarding what is done during visits.

Patient, and Getting Timely Help (which asks about hospice availability during evenings and holidays). These measures are all related to the frequency of visits made by a hospice.

It is also reported that the above CMS indicators correlate well with survey-based measures of caregivers and patients. In fact, hospices have complained that CMS does not track non-nursing visits from the professional team, which my dataset actually reports. Furthermore, it is reported that "Gaps between nursing visits has been an area of concern for the OIG".[14]

Note that while CMS collects quality data and family caregiver surveys, all the data collection began on or after 2017 and is often very sparse. Caregiver surveys are often not reported; other claim-based measures began from 2022 and may also be affected by COVID restrictions. Thus, they cannot be incorporated into my main analysis of dynamic competition.[15]

Finally, there has been a change in the Medicare hospice reimbursement scheme since 2017 where hospices are reimbursed for visiting patients in the last week of their life, suggesting CMS wants to incentivize greater visits during that period.

5. Medicare provides a checklist of "Suggested Questions To Ask When Choosing A Hospice". Many of these questions are related to frequency of visits – "What other members of the hospice team might I see, and how often will I see them?", "How many patients are assigned to each hospice nurse?", "Does the hospice have help after business hours? Nights? Weekends? Holidays?", "When I call with an urgent need, how long will it take for someone from the hospice team to respond", "How will the hospice team manage my pain or other symptoms that arise".[16]

6. I can compare my quality measure with those used for other providers in the literature. My measure aligns with the nursing home literature (Hackmann (2019), Lin (2015)), which measures nursing home quality by nurses-per-resident. Compared to the literature on other provider types, my measure has both advantages and disadvantages. The hospital literature uses readmission and mortality rates, while the dialysis literature uses infection rates (e.g. Grieco and Mcdevitt (2017)). Their advantage is that they measure outcomes; however, these are measures of extreme events – a hospital or dialysis center may provide poor quality yet not cause readmission, death, or infection in their patients. In contrast, my measure is

[14] These comments are from the following report by NHPCO: NHPCO report

[15] I merged the CAHPS family caregiver ratings of hospices for the year 2018 with my data and ran some descriptive regressions. The results are very noisy and rely on a small sample size, so they should be interpreted with caution. I find that visits-per-patient-day is positively but weakly partially-correlated with ratings. I also find that visits-per-patient-day is more strongly related to market share than ratings. The results are discussed and reported in Appendix M.

[16] Link to checklist: Medicare checklist

input-based, and so it has the advantage that it picks up more subtle differences in quality and effort.

Therefore, my quality measure reflects how often a hospice visits its patients. The dataset gives me total visits that a hospice makes to its entire pool of patients within a year and the number of patients it serves in the same year. The data also contains a breakdown of length-of-stay for the patient pool in a hospice-year (how many patients stayed for 0-7 days, 8-30 days, 31-90 days, 91-179 days, and 180+ days). This allows me to construct an "average length of stay" for patients at a given hospice-year. Table 3 shows the distribution of these variables.

I combine the above variables to construct my measure of quality "visits-per-patient-day" as follows. Multiplying the total patients at a hospice with its average length-of-stay gives the total patient-days at the hospice. Dividing total visits[17] in a year by a hospice with its total patient-days gives the visits-per-patient-day.[18] To deal with high outlier values, I censor the visits-per-patient-day variable at the 90th percentile. That is, for values which are greater than the 90th percentile, I recode those values to be equal to the 90th percentile value.

Table 3: Distribution of visits and enrollment

Variable	**10%**	**25%**	**50%**	**75%**	**90%**
Visits per patient	21.0	26.67	33.68	44.84	61.43
Average length-of-stay	35.91	42.28	49.82	58.19	66.54
Visits per patient-day	0.48	0.57	0.69	0.84	1.06
Patients	63.0	149.0	289.5	511.75	765.0
Market share	2.77	7.37	19.71	40.04	100.0

Notes: For this table, "market share" denotes within-hospice market share, i.e. fraction of hospice patients in a market-year enrolling into a hospice. Each observation is at the hospice-year level.

I conduct multiple empirical checks to strengthen my claim that visits-per-patient-day is a good proxy for effort and quality. First, I regress visits-per-patient-day on the number of rivals and other covariates, and show that greater competition is associated with a hospice increasing its visits-per-patient-day. This aligns with the idea that hospices are choosing visits-per-patient-day to compete for patients (Table 4). Second, I regress market share on visits-per-patient-day and other covariates,

[17] I aggregate all visits across staff type rather than select specific staff types, or allow different staff to contribute differently to the overall quality measure. This is because in my dataset, I see all staff types co-move very closely for hospices; see Section E.5 for Principal Components Analysis.

[18] It is often the case that patients need more visits near the end of their length of stay compared to earlier. Thus I construct the average length of stay to reflect this possibility. For the length-of-stay bins, I construct the chosen point in each bin to be 7, 25, 60, 100, and 230. Then the number of patients in each bin is multiplied with the corresponding point. The intuition is that patients in bins 0-7 days and 8-30 days need visits over most of their length of stay; however, longer staying patients may need visits for fewer days than their length of stay. I also estimate the demand system with visits-per-patient as the quality variable and obtain similar results.

Table 4: Regression of quality on level of competition.

	Quality			
	(1)	(2)	(3)	(4)
Count of rivals	0.037***	0.032***	0.029***	0.024*
	(0.005)	(0.008)	(0.007)	(0.009)
Count of rivals squared	-0.001***	-0.001**	-0.001**	-0.001*
	(0.000)	(0.000)	(0.000)	(0.000)
Reported average cost of visit			-0.140***	-0.141***
			(0.018)	(0.018)
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects				Yes
Controls		Yes	Yes	Yes
Obs.	2,066	2,066	2,066	2,066

Notes: Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators. Standard errors clustered at county-level in parentheses. Reported estimates and standard errors rounded to 3 decimal places.

and find that higher visits-per-patient-day is associated with higher market share (Table 5). This aligns with the idea that visits-per-patient-day is a measure of the quality that patients consider when choosing hospices.

I also study the persistence of quality choices over time, and how quality choices correlate with average lengths-of-stay. First, I find that there is notable dispersion in the quality choices made by hospices going from one period to the next; this is illustrated in Figure 1. Looking ahead, this dispersion will inform the estimates in Table 5, and will help with estimating the demand model. Second, I find no partial correlation between visits-per-patient-day and average length-of-stay; this is illustrated in Table 29 and Figure 4. This assuages worries that visits-per-patient-day is driven by selection on lengths-of-stay.

3.5 Quality-history of hospices

Next, I show suggestive evidence that hospices accumulate quality-history over time to gain market share. These will be based on two pieces of intuition. First, hospices which consistently chose high quality in the past have built up their quality-history, and so should have higher current market share. Second, new entrants should see their market share grow over time as they build up their quality-history. I find evidence for both in my dataset.

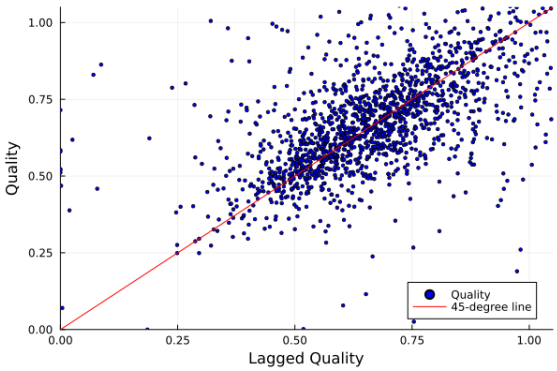
First, I show that past visits-per-patient-day impacts current market share. I regress a hospice's

Table 5: Regression of market share on quality and lagged quality.

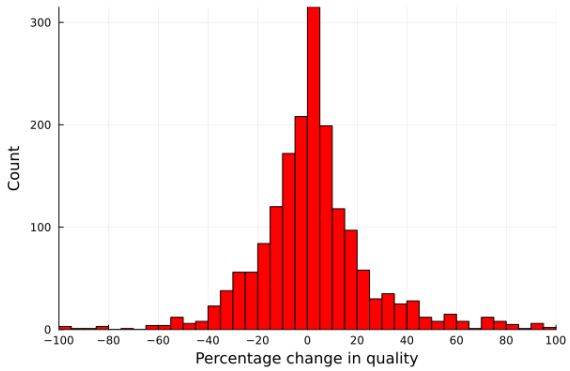
	log(market share)				
	(1)	(2)	(3)	(4)	(5)
Count of rivals	-0.058***	-0.051***	-0.038***	-0.033*	-0.045**
	(0.011)	(0.010)	(0.009)	(0.013)	(0.013)
Visits-per-patient-day	1.138***	0.666***	0.558*	0.445*	0.450**
	(0.253)	(0.173)	(0.223)	(0.203)	(0.150)
Visits-per-patient-day$_{t-1}$		0.410**	0.366**	0.322*	
		(0.126)	(0.127)	(0.139)	
Visits-per-patient-day$_{t-2}$		0.449**	0.361**	0.236	
		(0.136)	(0.122)	(0.146)	
Visits-per-patient-day$_{t-3}$			0.496***	0.449**	
			(0.120)	(0.135)	
Visits-per-patient-day$_{t-4}$			0.272**	0.228**	
			(0.088)	(0.080)	
Visits-per-patient-day$_{t-5}$			0.084	-0.034	
			(0.160)	(0.171)	
Visits-per-patient-day$_{t-6}$			-0.041	-0.167	
			(0.151)	(0.158)	
Sum of past 3 quality choices					0.314**
					(0.100)
County Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects				Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Obs.	2,066	1,597	937	937	1,403

Notes: The notation Visits-per-patient-day$_{t-n}$ denotes the visits-per-patient-day chosen n years prior. Sum of past 3 quality choices = $\sum_{i=1}^{3}$ Visits-per-patient-day$_{t-i}$. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators. Standard errors clustered at county-level in parentheses.

Figure 1: Variation in quality choice across periods.



(a) Scatterplot of current versus lagged quality (censored observations excluded).



(b) Histogram of percentage change in quality from one period to the next.

current market share on its current and past visits-per-patient-day, as well as other covariates (Table 5). I find that lags of visits-per-patient-day have statistically significant and positive impacts on current market share, and improve the overall fit of the regression. In addition, I find that quality choices further in the past have generally smaller explanatory power, and after 4 years have no explanatory power anymore. This suggests that the effect of quality choices further in the past declines over time, and thus informs my model setup. I also repeat the regression by instrumenting for current quality and stock of past qualities with current and lagged reported per-visit cost (as cost shifters), and find similar results (Table 30).

Second, I show that new entrants' market shares grow in the years following entry, then level out over time. This is suggestive of the importance of quality-history: if quality-history buildup helps a hospice attract patients, then new entrants should see growing market shares after entry, and that growth should slow down with time. I limit my sample to hospices which enter within my dataset's time coverage, then regress their market share on firm fixed-effects, indicator of entry year, and indicators for years after entry. Table 6 shows the results. With the firm fixed-effects picking up the average market share of each hospice in my dataset, the years of entry indicator variables measure how market share adjusts to the average over time. I find that the market share of a new entrant increases over time, rather than adjust to the average immediately. While part of the deficit in the first year could be because the hospice may not have operated for the entire year, the pattern is present for subsequent years as well.

3.6 Cost measures from hospice cost data

The data includes breakdown of annual hospice expenses. From this, for each hospice-year, I construct measures of per-period fixed cost, per-patient non-visit cost (hereafter referred to as

Table 6: Regression of market share on indicators for entry year and years after entry.

	log(market share)			
	(1)	(2)	(3)	(4)
Entry year	-0.916***	-0.942***	-0.952***	-0.960***
	(0.098)	(0.102)	(0.105)	(0.107)
Entry year + 1	-0.313***	-0.338***	-0.349***	-0.357***
	(0.067)	(0.070)	(0.072)	(0.078)
Entry year + 2		-0.110*	-0.120*	-0.128*
		(0.044)	(0.051)	(0.054)
Entry year + 3			-0.043	-0.051
			(0.040)	(0.044)
Entry year + 4				-0.040
				(0.046)
Firm Fixed Effects	Yes	Yes	Yes	Yes
Obs.	977	977	977	977

Notes: Standard errors clustered at county-level in parentheses. Entry year + n is an indicator variable for the year n after entry by a hospice.

"intercept cost"), and cost of making a visit (hereafter referred to as "visit cost"). All cost measures used in the analysis are inflation-adjusted. Details of variable construction, as well as a list of all the variables in the cost data, are listed in Appendix L.

4 Structural model

To study and model hospice competition, I set up a structural model of demand and supply for the hospice industry. The demand side is a discrete-choice model that incorporates a measure of quality-history of the hospice. The supply side is a dynamic oligopoly model where forward-looking hospices are choosing quality.

4.1 Demand

The demand for hospices is modeled as a nested-logit discrete choice model, with a nest on hospices. A consumer can choose between any hospice present in the market in that period, or an outside option. The outside option of not choosing a hospice can be interpreted as the consumer continuing curative treatment, or choosing to be cared for by family/nursing home.

The payoff to consumer i from choosing hospice j in period t is given by:

$$u_{ijt} = \alpha_{m(j)} + X'_{jt}\beta_d + \psi_{jt} + \xi_{jt} + \zeta_i + (1-\sigma_n)\tilde{\varepsilon}_{ijt} \tag{1}$$

$$\xi_{jt} = \rho\xi_{jt-1} + \varepsilon_{jt} \tag{2}$$

where $m(j)$ denotes the market in which hospice j is present and X_{jt} represents hospice characteristics[19]. The term ξ_{jt} denotes hospice-specific unobservables that are allowed to be serially correlated and vary over time. This can reflect factors like how connected the hospice is with physicians/nursing homes, advertising, past scandals, additional hospice services, etc. The ξ_{jt} are assumed to be orthogonal to firm characteristics X_{jt}, as is standard in the discrete-choice estimation literature. The term ψ_{jt} is a function of hospice j's current and past quality choices, explained below. Finally, $\zeta_i + (1-\sigma_n)\tilde{\varepsilon}_{ijt}$ reflect that this is a nested logit – there are two nests, one on all hospices and one on the outside option.

The term ψ_{jt} captures how a hospice's current and past quality choices affects consumer choice. Next I explain how I model ψ_{jt}.

Let a_{jt} be the visits-per-patient-day made by hospice j in period t. Let Γ_{jt} be the discounted sum of a hospice's past quality choices:

$$\Gamma_{jt} = a_{jt-1} + (1-\tau)a_{jt-2} + (1-\tau)^2 a_{jt-3} + \ldots = \sum_{i=1}^{age_{jt}} (1-\tau)^{i-1} a_{jt-i} \tag{3}$$

where τ is a depreciation parameter explained below. This implies the following transtion equation for Γ_{jt}:

$$\Gamma_{jt} = a_{jt-1} + (1-\tau)\Gamma_{jt-1} \tag{4}$$

I model ψ_{jt} as:

$$\begin{aligned} \psi_{jt} &= \eta_1 a_{jt} + \eta_2 \Gamma_{jt} \\ &= \eta_1 a_{jt} + \eta_2 \sum_{i=1}^{age_{jt}} (1-\tau)^{i-1} a_{jt-i} \\ &= \eta_1 a_{jt} + \eta_2 \left[a_{jt-1} + (1-\tau)a_{jt-2} + (1-\tau)^2 a_{jt-3} + \ldots \right] \end{aligned} \tag{5}$$

The term Γ_{jt} captures the idea of quality-history – past quality choices accumulate, which in

[19]The hospice characteristics are dummy variables for the following: hospice inpatient unit, pediatric program, bereavement services, day care for adults, for-profit status, free-standing hospice, home-health-based hospice, hospital-based hospice, and hospices classified as having agency-type "Other".

turn influences consumer choice in the current period.[20] The parameter η_2 captures the "quality-history effect" on consumer choice. The parameter τ can be thought of as a "quality-history depreciation parameter" that models how quality choices further in the past have a declining effect. Note that the impact of a firms' age is already subsumed in my measure of quality-history – a firm who has lived longer has accumulated a larger number of past quality choices into its current quality-history stock.

4.2 Supply

To understand how hospices choose quality over time, I set up a discrete-time infinite-horizon dynamic oligopoly model. To do so, I combine the above demand system with cost function and payoff function to formulate a strategic dynamic optimization problem. This broadly follows the framework of Ericson and Pakes (1995).

In the dynamic game a period is defined to be a year. For the rest of this section the time subscript is suppressed for clarity unless necessary.

4.2.1 Cost function and per-period profit

The cost to a hospice for serving a patient depends on its quality choice and other cost components. I impose the cost to be linear in visits-per-patient-day.

The expected cost of serving each patient at quality a_j (henceforth referred to as marginal cost) is given by:

$$MC_j(a_j) = \gamma_{0j} + \gamma_{1j} a_j \overline{LOS} \tag{6}$$

where $\overline{LOS}$ is the expected length-of-stay for a patient. Thus $a_j\overline{LOS}$ captures the average number of visits made to a patient over her entire length of stay, γ_{1j} is the cost of an additional visit, and γ_{0j} captures the non-visit cost for each patient. The functional form captures the intuition that marginal cost is linearly increasing in quality a_j given $\gamma_{1j} > 0$. From before, I refer to γ_{1j} as the per-visit cost and γ_{0j} as the intercept cost.

Combining the demand function with Medicare prices and market size, the per-period profit of hospice j in market m choosing quality a_j is:

$$\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) \left[P_m^{MCAR} \overline{LOS} - MC_j(a_j) \right] - \phi_j \tag{7}$$

[20] There are other ways of defining ψ_{jt}. One would be $\psi_{jt} = \eta a_{jt} + (1-\tau)\psi_{jt-1} = \eta \sum_{i=0}^{age_{jt}} (1-\tau)^i a_{jt-i}$. This is more parsimonious than my current expression but is also less data-intensive. I find that this yields similar demand estimates, see Table 27.

where $\mathbf{x}_m$ denotes the state variables in market m,[21] $\mathbf{a}_{-j}$ denotes the vector of realized actions of all rival firms in the market, M_m is the market size, and ϕ_j is the per-period fixed cost of hospice j. Medicare's per-day rate P^{MCAR} is multiplied with the expected length-of-stay $\overline{LOS}$, giving the expected revenue generated by a patient over their length-of-stay. This profit function is interpreted as maximizing the sum of expected profit per patient. This is because when choosing quality, a hospice does not know the exact realizations of length-of-stay for each patient it will enroll. In Appendix I.2 I derive this profit function and discuss alternatives.

For estimation, I impose additional restrictions on the cost parameters. I assume that γ_{0j} and ϕ_j varies by market ($\phi_j \equiv \phi_{m(j)}$ and $\gamma_{0j} \equiv \gamma_{0m(j)}$).

For the demand estimation I use the continuous measure of quality choice; however for estimating the supply side, I discretize quality choices into six tiers. This has the advantage of improving computational tractability (especially regarding counterfactuals) while still being a meaningful improvement of quality.

4.2.2 Dynamic choice of quality

The dynamic oligopoly game is modeled as a discrete-time simultaneous-move game over an infinite horizon. Every period (year) the incumbents decide what level of quality to provide. All consumers of a hospice receive the quality level chosen. The importance of quality-history for consumer choice means that current quality choices affect future sales, hence a forward-looking model is required; hospices also make this decision in a competitive setting, hence a model of strategic interaction is needed.

Hospice j's value function is:

$$V_j(\mathbf{x}_m, \varepsilon_j^a, \sigma; \theta) = \max_{a_j \in \mathscr{A}} \mathbb{E}\left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) + \varepsilon_j^a(a_j) + \beta V_j(\mathbf{x}'_m, \varepsilon_j'^a, \sigma; \theta) \middle| a_j, \mathbf{x}_m, \sigma \right] \quad (8)$$

where θ is the set of structural supply-side parameters, $\beta \in (0,1)$ is the per-period discount rate, ε_j^a is a vector of choice-specific errors for hospice j, $\varepsilon_j^a(a_j)$ is the choice-specific errors for hospice j choosing action a_j, and σ is the equilibrium strategy profile of the hospice and its rivals (to be defined later). The symbol $\mathscr{A}$ denotes the set of possible quality choices (recall that we discretize quality into six tiers for the supply side estimation), and a_j is the quality level chosen from this set. The choice-specific error terms are meant to capture unobservables affecting a firm's quality choice. This choice-specific structural error term is known to the firm but unknown to the econometrician. In keeping with the literature and to help with tractability in estimation, I assume these

[21] The state variables that comprise $\mathbf{x}_m$ are: all hospices' characteristics and demand shocks, all hospices' sum of discounted past visits, Medicare reimbursement rates, per-visit costs, fixed costs, intercept costs, market size, and identity of the county where it operates.

shocks to be *i.i.d* and distributed Type-1 Extreme Value.

The above setup reflects two assumptions about the choice-specific shocks widely used in the dynamic games literature. First, the choice-specific shocks enter the payoff function additively. Second, the choice-specific shocks evolve independently of other state variables.[22]

4.2.3 Equilibrium and Conditional Choice Probabilities

This section describes the equilibrium concept used and how it leads to the conditional choice probabilities (CCPs) predicted by the model. These CCPs will later be matched to observed probabilities to estimate the cost parameters. For this subsection, the notation for parameters θ is suppressed from the arguments for clarity.

I impose restrictions following the empirical literature on dynamic games. I focus on the Markov Perfect Equilibrium (MPE) of this dynamic game. This is a refinement of Nash Equilibrium which imposes that a player's strategy depends only on "payoff-relevant" state variables rather than on potentially the whole history of play. Intuitively, it limits the number of variables that a player looks at before taking an action. This introduces simplicity as well as ruling out a vast number of multiple equilibria. I also impose the policy functions to be symmetric and anonymous. The symmetry assumption imposes that all firms have the same policy function. The anonymity assumption imposes that a firm does not condition its choice on the identity of its rivals.

Each firm's strategy thus depends only on current period's payoff-relevant state variables and its private choice-specific shocks. Therefore a firm's strategy can be written as a mapping from current-period states and private choice-specific shocks to actions:

$$\sigma_j : \left(\mathbf{x}_m, \varepsilon_j^a\right) \to \mathscr{A}$$

Let the vector $\sigma = \{\sigma_j\}_{\forall j}$ denote the strategy profile of all firms in a market. I now show how this can be used to construct CCPs.

[22] Below I list these assumptions explicitly; the following mirrors the discussion in Lin (2015). Let the vector of common knowledge state variables in the current period be $\mathbf{x}$. The vector of common knowledge state variables and private information is denoted as $\mathbf{s} = (\mathbf{x}, \varepsilon^a)$. The state-to-state transition probability is $F(\mathbf{s}'|\mathbf{s}, \mathbf{a})$, where $\mathbf{a}$ denotes the vector of firm choices in the current period. Following Rust (1987), the two assumptions can be stated as:

1. Additive Separability: This imposes that the choice-specific structural error ε^a enters the per-period payoff function additively, i.e. $\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m, \varepsilon_j^a; \theta) = \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) + \varepsilon_j^a(a_j)$.

2. Conditional Independence: This comprises two assumptions. One, $\mathbf{x}'$ is independent of ε^a after conditioning on $(\mathbf{x}, \mathbf{a})$. Two, ε^a is *i.i.d.* and evolves independently of other state variables. Mathematically, this can be written as:

$$F(\mathbf{s}'|\mathbf{s}, \mathbf{a}) = F(\mathbf{x}', \varepsilon^{a'}|\mathbf{x}, \varepsilon^a, \mathbf{a}) = F(\mathbf{x}'|\mathbf{x}, \mathbf{a}) F_\varepsilon(\varepsilon^{a'})$$

where $F_\varepsilon(\cdot)$ denotes the distribution of ε^a.

The choice-specific value function of action a in state $\mathbf{x}_m$, $\hat{v}_j(a, \mathbf{x}_m, \sigma)$, is the discounted sum of payoffs to hospice j from choosing action a before choice-specific shocks ε_j^a are observed. Let the ex-ante value functions $V_j(\mathbf{x}_m, \sigma)$ be Equation (8) with the choice-specific errors integrated out:

$$V_j(\mathbf{x}_m, \sigma) \equiv \int V_j\left(\mathbf{x}_m, \sigma, \varepsilon_j^a\right) dG\left(\varepsilon_j^a\right) \tag{9}$$

The ex-ante value function can be written in terms of choice-specific value functions:

$$\begin{aligned} V_j(\mathbf{x}_m, \sigma) &= \int V_j\left(\mathbf{x}_m, \sigma, \varepsilon_j^a\right) dG\left(\varepsilon_j^a\right) \\ &= \int \max_{a_j \in \mathscr{A}} \left\{\hat{v}_j\left(a_j, \mathbf{x}_m, \sigma\right) + \varepsilon_j^a\left(a_j\right)\right\} dG\left(\varepsilon_j^a\right) \end{aligned} \tag{10}$$

The MPE is the strategy profile σ^* such that every firm is choosing the optimal strategy given the strategies of their rivals:

$$\sigma_j^*\left(\mathbf{x}_m, \varepsilon_j^a, \sigma^*\right) = \arg\max_{a_j \in \mathscr{A}} \left\{\hat{v}_j\left(a_j, \mathbf{x_m}, \sigma^*\right) + \varepsilon_j^a\left(a_j\right)\right\}$$

Given this optimal strategy profile, the CCP of an action a_j taken by firm j can be written as:

$$\Psi_j\left(a_j, \mathbf{x}_m, \sigma^*\right) = \frac{\exp\left\{\hat{v}_j\left(a_j, \mathbf{x}_m, \sigma^*\right)/\theta_e\right\}}{\sum_{a \in \mathscr{A}} \exp\left\{\hat{v}_j\left(a, \mathbf{x}_m, \sigma^*\right)/\theta_e\right\}} \tag{11}$$

where θ_e is the logit scaling parameter of the Type-1 Extreme Value logit error ε^a. Since the choice-specific structural error is known to the firm but unknown to the researcher, I integrate out the errors and recover the CCP. The CCP therefore gives us the choice probabilities of all available actions for a given observed state.

Under the assumption of logit error, we can write the choice-specific value function $\hat{v}(a_j, \mathbf{x}_m, \sigma^*)$ as:

$$\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*) = \sum_{\mathbf{a}_{-j} \in \mathscr{A}_{-j}} \left\{ \left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) + \beta V_j(\mathbf{x}'_m, \sigma^*)\right] F(\mathbf{x}'_m | \mathbf{x}_m, a_j, \mathbf{a}_{-j}) \prod_{n \neq j} \Psi_n(\mathbf{a}_{-j}[n], \mathbf{x}_m, \sigma^*) \right\} \tag{12}$$

where $\Psi_n(\cdot, \mathbf{x}_m, \sigma^*)$ is the CCP of firm n, $\mathscr{A}_{-j}$ is the set of all possible rival action profile vectors, and $\mathbf{a}_{-j}[n]$ is the action by firm n.

The logit assumption also allows us to rewrite Equation (10) as:

$$V_j(\mathbf{x}_m, \sigma^*) \quad = \quad \theta_e \left[0.577216 + \ln\Big(\sum_{a_j \in \mathscr{A}} e^{\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*)/\theta_e}\Big)\right] \tag{13}$$

4.3 Timing

In any period, the industry state comprises common-knowledge variables (own characteristics, rival characteristics, market characteristics) and private information (choice-specific structural errors ε^a). The industry state comprises both endogenous and exogenous variables, and as a result evolves due to a mix of firm decisions and exogenous transitions.

Given the model above, the game's evolution and timing is as follows. For period t and market m:

1. Incumbents observe $\mathbf{x}_{mt}$ and structural errors, and each make quality choices.
2. Quality-history Γ_{jt} of each incumbent evolves.
3. Consumers observe $\mathbf{x}_{mt}$, quality-histories, and structural errors, then choose a hospice.
4. Market shares are realized, and incumbents receive profits.
5. Market structure evolves exogenously due to entry and exit.
6. All state variables evolve.

4.4 Discussion

Consumer payoff versus utility, and interpretation of quality in demand: Note that I call the consumer's discrete choice function a "payoff" rather than "utility". This is because the function includes quality-history and possibly imperfect information on current quality. It is unclear how these would affect a consumer's experienced utility and satisfaction. Quality-history mainly serves to influence a consumer's choice rather than their experience (that said, consumers may experience greater satisfaction from choosing a hospice with better quality-history, but the welfare implications of that are unclear). Regarding the inclusion of current quality, since the consumer has limited information on it, this only serves to capture how any limited information on current quality can sway consumer choice rather than how it affects underlying utility.

Thus, this consumer payoff function aims to capture the factors that influence consumer choice rather than model the underlying consumer experience and welfare. By not calling it "utility function", I am avoiding giving the impression that quality-history is giving consumers "utility" in an ad-hoc manner.

A similar decision is made in Abi-Rafeh et al. (2026) and Dubois et al. (2018) – both include advertising stock in the function and thus face the same problem about interpreting welfare and utility. They call their functions "payoff" (Dubois et al. (2018)) and "decision utility" (Abi-Rafeh et al. (2026)).

On a related note, I now discuss how to interpret the quality coefficient in the demand model. Since information on current quality a_{jt} is limited, the term $\eta_1 a_{jt}$ is not a measure of how much consumers value quality, but rather how much limited information on current quality sways observed consumer choice. Thus, the coefficient η_1 is not a measure of consumer valuation of true current quality.

Identification of quality valuation in demand models: Papers studying consumer choice of healthcare provider usually project market share on current quality and other provider attributes, and use that to understand how much patients value quality. How does the omitted quality-history affect such demand models? Quality-history may be subsumed in the unobservables in the demand model; or, if past and current quality choices are highly persistent, then some of their effect may be subsumed in the valuation of current quality. This suggests that the data variation being used to estimate quality valuation in demand models without accounting for quality-history may be problematic. Furthermore, the quality valuation in demand models (alongside other product attributes) is used to estimate the cost function of the firm (particularly the cost of providing higher quality), which is then used to perform counterfactuals. If the quality coefficient is not estimated off of the right variation, then it is unclear how accurate the cost function is as well.

Modeling the effect of quality-history: I approximate the effect of quality-history on consumer choice as a parametric function of its past quality choices. This parametric approximation allows me to be agnostic about the specific channel through which quality-history operates. Consumers may care about quality-history because of the corresponding goodwill and name recognition, or because quality-history helps predict current quality in the face of bounded rationality, or consumers care about how long a hospice has serviced a market with reasonable quality. It also allows me to study the effect of quality-history using only firm-level data and maintain computational tractability.

Another element of the model is that past quality inside Γ_{jt} depreciates at the rate of τ. There are several reasons behind incorporating this depreciation element:

1. Information on past quality choices disappears over time. This might be because of forgetting by the community. It could also be due to retirement or relocation of physicians, social workers, and community members. The patient's family is less likely to require a hospice soon after the patient passes away, so the fact that there are no repeat consumers means information is less likely to persist. Another way of thinking about this is that information on past choices is still available but gets more costly to retrieve over time.

2. Past quality choices may be less relevant over time. This is because past quality choice also

represents old and outdated medical practice styles, technology, and medicine. An example is that a hospice nurse's visit 20 years ago had very different content compared to a hospice nurse's visit at present. Quality choices very far in the past reflect the hospice's demand shocks, costs, and characteristics that may no longer have any predictive power, since these are also changing over time.

Finally, an important advantage of measuring quality-history as a single statistic of discounted past quality choices is that it makes the dynamic game estimation and solution tractable. Other functional forms (such as a moving average or a flexible function over several lags of quality) would have required me to track the exact sequence of quality choices for each hospice in each market. This would make the state space extremely large and the resulting dynamic game intractable. Instead, modeling quality-history as discounted past quality choices Γ_{jt} helps, because it evolves in a deterministic way for a given quality choice: $\Gamma_{jt} = a_{jt-1} + (1-\tau)\Gamma_{jt-1}$. In the dynamic game, so as long as firms track Γ_{jt}, they can predict how it will evolve conditional on the current action of firms; they no longer need to track the entire history. This is a contribution of my paper – I incorporate the history of quality choices as a dynamic state in a tractable way through this statistic. As a result, I can study the Markov Perfect Equilibrium of this game.

Presence of current quality in the demand system: Recall that I include current quality in the demand system. This is for several reasons. Consumers may be able to partially observe current quality through visiting/conversing with the hospice and current patients. For instance, a hospice might be able (to some extent) to convince a potential consumer about the level of quality they will provide. Furthermore, my data is at the yearly level, so information on current quality has time to diffuse through the community over the course of a year. This specification also lets the data flexibly determine how much weight to put on current versus past qualities.[23]

Additional discussion: Detailed discussions on modeling choices and estimation are presented in the Appendices for space limitations. Below I list the key Appendices and summarize their contents.

In Appendix G.1, I discuss the persistent unobserved demand shocks in the demand model, its estimation (including its AR(1) structure), and its interpretation within my model. I think of it as capturing non-quality dimensions such as connectedness with physicians, scandals, factors outside the hospice's control, etc.

[23] My estimation could have found η_1 to be zero, suggesting current quality has no effect. I also estimate demand where I fix η_1 to be 0 and see how much η_2 changes, and it does not change by much (Table 26). It subsumes models of quality competition in most empirical papers that include only current period quality in the demand system.

See Appendix G.2 for a discussion on capacity constraints and why it is unlikely to play a role in my setting; Appendix G.3 for a heuristic discussion on identification of demand and cost parameters; and Appendix G.4 for discussions on advertising and additional dimensions of Medicare hospice reimbursement.

5 Estimation and Results

This section details how the structural model above is estimated and the estimation results. The demand model is estimated using GMM following the inversion technique of Berry (1994). The supply side is estimated by leveraging additional cost data and using the methods of Bajari et al. (2007) and Pakes et al. (2007) that match simulated choices with observed choices. Demand estimates show that consumers are more likely to choose a hospice with better quality-history, and the effect of past quality choices depreciates slowly each year. Supply side estimates show that an additional visit by a hospice to a patient costs ranges from $153 to $256, depending on the hospice and the county it is located in.

5.1 Demand estimation

Using Berry (1994), the nested-logit discrete choice model can be rewritten in terms of observed market shares:

$$ln(s_{jt}) - ln(s_{0m(j)t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} + \eta_1 a_{jt} + \eta_2 \Gamma_{jt} \tag{14}$$

where s_{jt} and $s_{0m(j)t}$ represent the market shares of hospice j and the outside option respectively, and $s_{j|gt}$ is j's within-hospice market share. Recall that for flexibility we also allow the unobserved demand shocks to be persistent, i.e., $\xi_{jt} = \rho\xi_{jt-1} + \varepsilon_{jt}$.

These equations are estimated simultaneously via GMM. Instruments are multiplied with the error term ε_{jt} to give moment conditions for GMM.[24] Further details of demand estimation are given in Appendix H.1.

Next I discuss endogeneity concerns and choice of instruments. Within-hospice market share

[24] Interacting instruments with ε_{jt} instead of ξ_{jt} also allows for the possibility that a hospice with high ξ might face less entry because it is more difficult to compete against. Without it, I would have to assume that level of competition in the market is orthogonal to the ξ of all incumbents. But that might not be true: if the incumbent has persistently high (or low) ξ, then that could affect entry/exit decisions and hence the level of competition in the market. By allowing for persistence, the claim is that level of competition is orthogonal to some residual term after accounting for persistence. In the demand estimation this means I build moments around ε_{jt} instead of ξ_{jt}; this follows Grennan (2013) and Hodgson (2023).

$s_{j|gt}$ is mechanically correlated with unobserved demand shock ξ, and it may be that quality-history is also correlated with ξ. As a result, I use instruments that are orthogonal to ε_{jt} in the ξ transition equation. My core set of instruments come from Berry et al. (1995) (henceforth BLP IVs), reported per-visit costs from hospice cost data, and Medicare's reimbursement rates.[25][26] Furthermore, I use lagged number of rivals and lagged reported per-visit costs as additional instruments for Γ_{jt}.

Table 7 shows the results from the demand estimation. The quality-history depreciation parameter τ is estimated to be 0.25, which suggests past quality choices have highly persistent effects. Positive values of η_1 and η_2 mean that a hospice with higher current and past quality choices is more likely to be chosen by consumers; in other words, a hospice with better quality-history will have higher market share. The unobserved demand shocks are highly persistent (ρ), and consumer preferences within a nest are highly correlated (σ_n). The remaining values in Table 7 gives consumer preferences for hospice characteristics.

To aid interpretation of the estimates of η_1 and η_2, Table 8 shows the distribution of elasticities with respect to Γ_{jt} and a_{jt} across all hospice-year observations. The elasticity with respect to sum of discounted past visits is generally higher compared to that of current quality.

I conduct additional robustness checks for demand estimation, involving Gandhi-Houde IVs (Gandhi and Houde (2019)), county-dummy-by-fuel-price as cost shifters, subsets of instruments, not imposing the AR(1) structure on ξ_{jt}, etc. All specifications yield the same qualitative results; see Appendix C.

5.2 Cost function from cost data

From the cost data I construct hospice-by-year estimates of per-period fixed cost, intercept cost, and per-visit cost (adjusted for inflation rates). I regress each on county dummies to obtain county-

[25] BLP IVs are standard instruments in the IO literature that use measures of competitiveness in product space to create variation in within-hospice market share and quality choice. In my setting, these are the sum of rivals and sums of rival characteristics. These BLP IVs shift quality choice by varying the level of competition that a hospice faces. The reported per-visit cost acts as a cost shifter. Medicare prices impact quality choice (see counterfactuals) while being orthogonal to firm-specific demand shocks since they are set at the national level and vary by county based on Medicare cost indices. Thus they vary quality by making patients more/less profitable to serve or by indicating potentially higher costs of providing care. I perform counterfactuals by using a variety of instruments and continue to obtain similar results. See Berry and Haile (2014) for the importance of cost-shifters in random coefficients discrete choice model estimation.

[26] There are several points to note about using per-visit costs as cost-shifters. First, I find in a separate paper (Alam (2025)) that firms do not reduce their per-visit costs in response to greater competition. Second, while I use reported per-visit costs as instruments for the demand side, I estimate separate per-visit costs for the supply side, with the idea that the estimated costs are "economic costs". I do find that economic costs are higher than the reported costs. By using the reported costs as IVs, I am arguing that the variation in reported per-visit costs across firms is informative about their underlying variation in costs, even though the levels of the reported per-visit costs may be lower than economic costs. In reduced-form work, I do find that firms with higher reported per-visit costs make fewer visits, which lines up with economic theory. This is further evidence that reported per-visit costs are informative about cost-differences across firms.

Table 7: Results of demand estimation.

Key Parameters			Hospice Characteristics		
	Estimate	(SE)		Estimate	(SE)
τ	0.247	(0.097)	For-profit	-0.039	(0.106)
η_1	0.505	(0.249)	Hospice inpatient unit	0.156	(0.124)
η_2	0.276	(0.111)	Pediatric program	0.192	(0.120)
σ_n	0.661	(0.107)	Bereavement services	-0.069	(0.021)
ρ	0.680	(0.115)	Day care for adults	0.092	(0.123)
			Agency type: free-standing	-0.050	(0.116)
			Agency type: home health based	-0.043	(0.112)
			Agency type: hospital-based	0.020	(0.152)

Notes: The demand model is estimated using GMM with BLP IVs, reported per-visit costs, and inflation-adjusted Medicare rates. All specifications include lagged rival count and lagged reported per visit cost as IVs. Bootstrapped standard errors clustered by county are in parentheses. Estimates of county fixed effects are suppressed.

Table 8: Elasticity estimates

Elasticity	**10%**	**25%**	**50%**	**75%**	**90%**
Γ_{jt}	0.222	0.642	1.237	1.748	2.103
a_{jt}	0.376	0.649	0.842	1.081	1.312

level averages. Table 9 shows the distribution of regression coefficients, while Table 43 shows the full regression results.

Table 9: Distribution of coefficients from regression of cost on county dummies.

	10%	**25%**	**50%**	**75%**	**90%**
Fixed Cost	213,449.68	360,520.09	534,721.52	919,459.91	1,142,199.24
Intercept	257.40	372.09	587.18	794.80	993.89
Per-visit cost (reported)	85.82	92.37	109.65	131.13	143.21

Notes: Detailed results are presented in Table 43.

5.3 Estimation of dynamic oligopoly model

As an alternative to using the per-visit cost from the cost data, I recover both the per-visit cost estimate and the logit scaling parameter for the hospice's quality choice using a two-stage estimator.[27] I combine the forward-simulation approach of Bajari et al. (2007) with the method-of-moments approach of Pakes et al. (2007); a similar implementation was done by Wang (2022). The first stage involves getting reduced-form estimates of the firms' policy functions from the data, as well as state transition probabilities. In the second stage, these policy functions are used to conduct forward simulation and generate model-predicted conditional choice probabilities (CCPs, explained in Section 4.2.3) for a given guess of cost parameters. Moment conditions are then constructed, and an optimizer searches to find cost parameter values that minimize the distance between model-predicted CCPs and observed probabilities.

To simplify computation and model solution, I discretize quality choice into 6 tiers. Visits-per-patient-day is discretized into 6 bins between 0.44 to 1.04 visits-per-patient-day (open lower intervals and closed upper intervals) with increments of 0.12. From Table 3 it can be seen that this approximately covers the 10th to 90th percentiles of the observed visits-per-patient-day distribution.

I do not estimate the per-period fixed cost and intercept cost in the dynamic game estimation, instead fixing them at the estimated values from the cost data (see Tables 9 and 43). These are allowed to vary by county; following the notation from earlier, $\phi_j \equiv \phi_{m(j)}$ and $\gamma_{0j} \equiv \gamma_{0m(j)}$, where $m(j)$ is the county where hospice j is located.[28]

Following Bajari et al. (2007), the first stage requires an estimate of the policy function from the data. Broadly speaking, I regress quality on a variety of covariates to see how quality choice is affected by state variables. This sheds light on how hospices react to quality-history and product characteristics of themselves and rivals. I interpret this regression as reflecting the policy function used by hospices when making quality decisions.

More specifically, I obtain an estimate of the policy function in two different ways. For the baseline case, I use a 2SLS regression with county fixed effects and multiple instruments (Table 16) to account for potential serially correlated unobservables following the ideas of Berry and Compiani (2021). As an alternative, I use a flexible regression (using second-degree polynomials

[27] There are two points to note about this. First, the reason I use a two-stage estimator instead of a full-solution method is because of the presence of a large number of state variables (see below). This renders a full-solution method computationally infeasible. Second, a two-stage estimator assumes that a single equilibrium is played across all markets, and so cannot allow for multiple equilibria. Fortunately this seems to not be an issue in my setting, because when solving the full model in counterfactuals I have been unable to find multiple equilibria despite extensive searching (although I have not formally proved uniqueness of equilibria); see Appendix H.5 for more details.

[28] Per-period fixed cost is not identified since I do not model exit. I also found the intercept cost to be hard to pin down during cost estimation.

of state variables) to approximate the policy function (Table 20).[29] I also conduct robustness check by including firm-specific fixed effects in the empirical policy function and obtain similar results; see Appendix B.4. See Appendix B for details on how the empirical policy function is estimated, the results of the estimation, and first-stage results for the 2SLS regression.

The state variables that enter a hospice's value function are its hospice characteristics and demand shocks, its rivals' characteristics and demand shocks, its sum of discounted past visits (Γ_{jt}), its rivals' sum of discounted past visits (Γ_{-jt}), Medicare reimbursement rates, per-visit costs, fixed costs, intercept costs, market size, and identity of the county where it operates.

Next I explain how I implement state transitions in my forward simulation. For the estimation, I assume for simplicity that a firm's characteristics do not change over time; this assumption can be relaxed. Firms' current period unobserved shocks (ξ) are imputed from the demand estimation; the estimated persistence rate is used to simulate it forward while drawing the errors (ε) from the econometric error distribution. Market size and Medicare rates follow an AR(1) process that is estimated from the data. From my data I find the average length-of-stay $\overline{LOS}$ to be 49.16 days, and I set the discount rate $\beta = 0.95$.

Entry is moderate and exit infrequent in my data, so I implement entry and exit as exogenous transitions rather than endogenizing them as part of the firm's choice problem. In my forward simulation, entry by potential entrants into a market is modeled through an ordered logit specified in Table 21. Exit by a firm happens with a probability of 4% every period, which matches exit rates in my data.

The first-stage policy functions and transition matrices are used in forward simulation to construct choice-specific value functions and moment conditions. This combines methods by Bajari et al. (2007) and Pakes et al. (2007). See Appendix H.2 for a description of the estimation procedure and Appendix A for an explanation of how I impute quality-history stocks of older hospices.

The simulated path continues for 80 periods. For each action and each observed state I run the forward simulation 200 times. The resulting choice-specific value functions are then used to construct the model-predicted choice probabilities.

Next, I describe how these model-predicted choice probabilities are used to estimate the hospice cost function. The intuition is that I search for values of parameters θ that minimize the distance between model-predicted choice probabilities and observed choices.

The dynamic oligopoly model is estimated using GMM. The GMM residual term for obser-

[29] While I track a wide variety of state variables, I use some aggregate measures of competition in the policy function due to data limitations. To account for competitive effects from rivals, I include the following state variables for a given hospice: number of rivals, number of rivals squared, difference between mean Γ_{jt} of rivals and own, difference between mean $X'_{jt}\beta_d$ of rivals and own, and difference between mean ξ_{jt} of rivals and own.

For the 2SLS empirical policy functions, I use $X'_{jt-1}\beta_d$ and ξ_{jt-1} as instruments for "mean Γ_{jt} of rivals - own Γ_{jt}". I find that the signs on the variables in the empirical policy functions as well as the first stage of the instruments on the dynamic states match the directions predicted by a fully-solved policy function.

vation n is $\Xi_n(\theta) = a_n^{data} - \sum_{a_n \in \mathscr{A}} a_n \hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$, where $\hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$ is the predicted choice probability for action a_n, a_n^{data} is the observed quality choice for observation n, θ is the guess of structural parameters, and $\mathbf{x}_n^{data}$ is the market state for observation n. The error term measures how close the average model-predicted choice is to the observed choice. The instruments Z used in GMM are the county indicators and the variables in the first-stage empirical policy functions, giving us the moment conditions $E[Z'\Xi_n(\theta)]$. The intuition for this is that the model's predicted choices leverage these variables, and so at the true parameter values the model's prediction errors should be orthogonal to these variables. The resulting GMM optimization problem is:

$$\min_{\theta} \left[\frac{1}{N}\sum_n Z_n'\Xi_n(\theta)\right]' \hat{W} \left[\frac{1}{N}\sum_n Z_n'\Xi_n(\theta)\right]$$

This objective function is minimized to estimate cost parameters of hospices (γ_{1j} and θ_e). Standard errors are calculated via block-bootstrapping, similar to Wang (2022).

Since there is heterogeneity in the reported per-visit costs, I allow the per-visit cost parameter γ_{1j} to also have some heterogeneity. I use two different specifications for this. In the first specification, I use the reported per-visit cost for each hospice from the cost data, scaled up by a common factor, i.e. $\gamma_{1j} = \gamma_{scale} \cdot \text{reported per-visit cost}_j$. I then estimate γ_{scale}.[30] In the second specification, I divide all counties into whether their mean reported per-visit cost is above or below the median, and allow the cost parameter to differ between these two groups, i.e. $\gamma_{1j} = \bar{\gamma} + \gamma_{11} \cdot \mathbf{1}\{m(j) \text{ is below median}\}$. I then estimate $\bar{\gamma}$ and γ_{11}.

Table 10 gives the results of the dynamic supply estimation. For the first specification, the scaling factor is estimated to be 1.79, which means that the cost of an additional visit is around 1.79 times the reported per-visit cost from the cost data.[31] Looking at the 10th and 90th percentiles of reported per-visit costs in my dataset, this scaling factor suggests estimated per-visit costs ranging from \$153 to \$256. For the second specification, the cost of an additional visit is around \$240 for high-cost counties and \$80 lower for low-cost counties.

I now give a brief discussion of the estimate of γ_{1j}. This estimate is higher than my measure of per-visit cost from the cost data (see Table 9), which had a median of \$109.65. The higher value from the dynamic game cost estimation is reasonable because it measure the economic cost, while the cost measures from expense data reflect accounting cost. Economic cost includes, in addition to accounting cost, factors such as opportunity cost, presence of constraints, difficulty in

[30] The fact that per-visit cost can vary by hospice is the sort of firm-level heterogeneity that makes quality-history particularly important for consumers to track – hospices with repeated high-quality choices are more likely to have cost advantages, and thus continue choosing high-quality.

[31] I use the reported per-visit cost for each hospice in each period. While I could average over all observations for a hospice and use its mean per-period reported cost, I find the autocorrelation between current and lagged reported cost to be extremely high (around 0.95 for my sample). Thus I expect it to make little difference to my estimation results.

hiring nurses, etc. In my policy counterfactuals I will therefore use the cost measure from the cost estimation.[32] Note that this paper's aim is to estimate a reasonable approximation of the cost function to model hospice quality competition; a separate paper (Alam (2025)) is focusing on hospice cost structure and efficiency differences using accounting cost reports.

Table 10: Results of supply estimation.

	(1)	(2)	(3)
γ_{scale}	1.792		
	(0.049)		
θ_e	15.139	22.763	42.307
	(1.542)	(5.132)	(14.374)
$\bar{\gamma}$		234.291	246.734
		(9.719)	(12.770)
γ_{11}		-76.869	-88.966
		(14.566)	(22.882)

Notes: The first column uses the per-visit cost function specification $\gamma_{1j} = \gamma_{scale} \cdot$ reported per-visit cost$_j$. The second and third column use the per-visit cost function specification $\gamma_{1j} = \bar{\gamma} + \gamma_{11} \cdot \mathbf{1}\{m(j) \text{ is below median}\}$. Each column uses a different empirical policy function. Specifications for empirical policy functions vary as follows. (1): Uses county fixed effects, instruments, reported per-visit costs, and other covariates; (2): Uses county fixed effects, instruments, and other covariates; (3): Uses county fixed effects and quadratics of variables (no instruments). All empirical poicy functions use covariates that convey information about level of competition, own quality-history, and rival quality-history. See Table 16 for the empirical policy functions used in first two columns, Table 20 for empirical policy function used in last column. Per-visit costs and logit scaling parameter θ_e are estimated using the methods of Bajari et al. (2007) and Pakes et al. (2007). Standard errors (in parentheses) are calculated via bootstrapping at the county-year level. 200 bootstrap runs were used. The logit scaling scaling parameter θ_e and its standard error are scaled by 10^5 for presentation purposes.

6 Policy counterfactuals

Given the above estimates of the structural parameters, I can now conduct counterfactuals to investigate the effects of reimbursement-based and information-based policy tools.

[32] I also interpret my cost estimates as subsuming any altruism by the hospice. That is, greater altruism by a hospice will lead to lower cost estimates – the hospices will be choosing higher quality than would be suggested by their accounting costs, and my estimation will rationalize this as the hospice having lower cost. See Hackmann (2019) for a paper that backs out the altruism parameter from the preference function.

6.1 Solving for equilibrium outcomes

To conduct counterfactuals I need to solve the model and recover the policy function of the hospice. This requires me to calculate $\Psi(a_j, \mathbf{x}, \boldsymbol{\sigma}^*)$, $\hat{v}(a_j, \mathbf{x}, \boldsymbol{\sigma}^*)$ and $V(\mathbf{x}, \boldsymbol{\sigma}^*)$ such that Equations (11), (12), and (13) simultaneously hold at every point in the state space. I solve the model using the policy iteration method (Rust (1994), Judd (1998), Benkard (2004)). This gives me the policy function and ex-ante value function at every action-state combination; thus, I solve for the optimal policy function and value function rather than using the empirical estimates from the data. To simplify computation, I also discretize the sum of discounted past visits Γ_{jt}, and impose a stochastic evolution process to minimize the effects of any such discretization (following Benkard (2004)). For more details on model solution, see Appendix H.4.

It is not computationally feasible to solve the model with the full set of state variables and multiple dynamic decisions. Thus, to ease computational burden, I solve a simpler version of my setting. First, I solve the model for a representative market instead of for each market in my dataset. Second, I limit the number of firms to 3 (unless stated otherwise). This is a good approximation of the competitive environment in my data, since firm-count at the market-year level has a median of 2. Third, I keep the number of firms fixed, thereby ruling out entry and exit. This is a reasonable assumption to make because entry happens in less than 10% of market-year observations, and exit is infrequent in the data.[33]

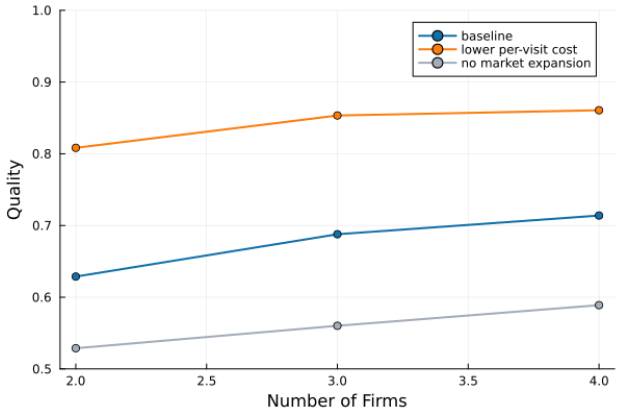
While not explicitly modeling entry or exit, I report the ex-ante value functions of incumbents for all policy simulations. Changes in ex-ante value functions can be indicative of entry-exit incentives; for instance, a policy change that lowers ex-ante value functions may induce exit, while an increase may incentivize entry. In Appendix I.1 I discuss the intuition behind this approach as well as what it rules out.

Counterfactual results are reported as follows. For a given policy experiment, I solve the model to obtain the policy function. The policy function gives the probability of a firm making each action choice for any given state under the chosen policy experiment. I pick a starting point for a market, then simulate forward 1000 periods using the policy function. I report the average quality choice by one firm over the simulation. To prevent my choice of initial conditions from contaminating the result, the first 200 periods are dropped; additionally I check by visual inspection to ensure that the industry has settled into its long-run state.[34]

[33] This rules out firms improving quality-history for entry deterrence. This is a reasonable assumption to make because I do not find strong evidence of entry deterrence through quality-history in my data – I regress entry numbers on quality-history of incumbents and do not find a relationship.

[34] While I show how different policy levers can affect equilibrium quality, I do not specify what the quality target should be. This is because my demand model is silent about consumer valuation of visits. The thought experiment is that CMS has figured out the quality target through studies of their own, and can achieve said target with the policy instruments I explore.

Figure 2: Equilibrium quality vs number of firms.



Notes: This graph shows the simulated equilibrium quality choice of a representative firm as the number of firms in the market increases, for different parameter values and economic environments. The baseline case is for a market with three identical firms. The other cases are for a market with all low-cost firms, and a market with no market-expansion effects.

Before conducting my main set of policy counterfactuals, I solve the model for different parameter values and economic environments to illustrate some key model characteristics. These are shown in Figure 2, and the findings are summarized below. Equilibrium quality rises as the level of competition increases, is higher when hospices have lower per-visit costs, and is lower when there are no market expansion effects. The latter shows how an additional incentive for increasing quality is to take market share away from the outside option, in addition to business-stealing from rivals.

6.2 Reimbursement-based policy counterfactuals

Increasing Medicare reimbursement rates: To see how hospices react to lower reimbursement rates, I study how hospice quality changes as Medicare's hospice reimbursement rate changes. This is motivated by the fact that hospices (and other healthcare providers) have frequently complained Medicare reimbursement rates are too low and have not kept up with inflation rates.

I solve the model for three identical firms under different Medicare reimbursement rates (Table 11). I find that higher Medicare reimbursement rates leads to higher equilibrium quality choice. The intuition is that higher rates means each patient is now more financially valuable to the hospice, so it is optimal to choose higher quality (and incur higher costs) to attract more patients.

I also find that market share and Medicare spending rises, with the latter rising more than proportionately to the increase in Medicare rates. This is once again due to market expansion effects – higher Medicare rates lead to higher equilibrium quality, which in turn leads to more consumers switching from the outside option to hospices. The increase in number of hospice

Table 11: Change in equilibrium quality with Medicare rates.

Per-Diem Rate	Quality	Market Share	Ex-Ante Value Function (normalized)	Medicare Spending (normalized)	Per-Diem Rate (normalized)
203.4	0.611	0.198	1.0	1.0	1.0
223.74	0.644	0.199	1.149	1.105	1.1
244.08	0.688	0.204	1.29	1.24	1.2
264.42	0.738	0.215	1.419	1.413	1.3

Notes: The "normalized" variables are normalized by their values at the first row.

patients then leads to more Medicare spending, which is amplified by the higher Medicare rates. This effect of market expansion mirrors the findings of Hackmann (2019) for the nursing home industry.

Lower Medicare reimbursement rates can also influence equilibrium quality choice by affecting entry and exit incentives. Table 11 also shows how the mean ex-ante value function of a hospice over the simulation changes as Medicare rate increases. Higher (lower) Medicare rate leads to higher (lower) ex-ante value function, which may lead to entry (exit). The change in the number of firms therefore leads to an additional impact on quality. In other words, higher Medicare rates can lead to higher quality both through more intense competition between existing firms as well as through entry of new firms.

Tying Medicare reimbursement to quality choice: Next I investigate an alternative reimbursement scheme for Medicare. The current reimbursement scheme involves Medicare paying hospices for every day a patient is enrolled, i.e. a per-day per-patient scheme. If we want to achieve a target quality level at the lowest possible Medicare spending, a natural route to investigate is to tie part of the reimbursement to quality.

To that end, I solve the model for hybrid per-day per-visit schemes, with the results shown for a duopoly in Table 12 and triopoly in Table 42. I progressively reduce the per-day rate and increase the per-visit rate while ensuring that under the new terms, the equilibrium quality choice stays the same. The total Medicare spending is calculated for each scheme; for clarity, I normalize the spending at each scheme by the total Medicare spending at the per-day per-patient rate (i.e. Medicare's current scheme).

I find that Medicare can reduce per-day rate and raise per-visit rate in such a way that total spending declines even as equilibrium quality choice stays the same. The intuition is that this hybrid scheme better aligns quality incentives for hospices. In the per-diem scheme, hospices would only raise quality in response to competition from other hospices and the outside option. The payoff from an additional visit is that the hospice can gain or defend market share. Under the

Table 12: Reimbursement schemes that lead to 0.627 visits-per-patient-day in a duopoly.

Per-Diem Rate	Per-day rate	Ex-Ante Value Function (normalized)	Medicare Spending (normalized)
244.08	0.0	1.0	1.0
207.47	30.0	0.866	0.924146
160.69	70.0	0.7	0.834388
114.51	110.0	0.539	0.745019
67.53	150.0	0.373	0.637597

Notes: The rates are in US dollars. The total Medicare spending associated with each scheme is normalized with respect to the spending of the per-day scheme. The mean ex-ante value function is normalized with respect to the ex-ante value function of the per-day scheme. As we move down rows, equilibrium quality stays the same yet Medicare spending declines.

hybrid scheme, hospices now get an additional payoff from visiting a patient through the per-visit reimbursement, causing equilibrium quality to be higher.

The takeaway is that potential cost savings can be achieved by shifting weight from per-day to per-visit reimbursement. Note that how far we can increase the per-visit rate is limited by the estimated marginal cost, since a per-visit rate exceeding the marginal cost will lead to hospices making unlimited visits to maximize profits.[35]

There is a trade-off associated with this scheme however – as we change the reimbursement structure in this direction, the average ex-ante value function of each hospice decreases, indicating that hospices are making lower profits. As a result, this scheme may risk inducing exit among hospices. Therefore, when implementing this scheme, policymakers have to be wary of not inducing exit and reducing competition.

Finally, instead of thinking of this counterfactual as tying hospice reimbursement to visits specifically, we can also interpret it as tying reimbursement to some measure of hospice quality. This measure of quality could be obtained via a comprehensive hospice-quality index constructed by CMS or through Family Caregiver surveys conducted every few weeks.

Some real-world evidence about the promise of such a hybrid-scheme is as follows. In 2016, Medicare introduced a "service intensity add-on" (SIA), which provides additional reimbursement to hospices for visits in the last week of life. While it does not quite match my hybrid scheme, it can provide some suggestive evidence about whether tying reimbursement to visits can increase visits. Christian et al. (2022) uses detailed data to find that nursing minutes went up in the last weeks of

[35] An alternative presentation of this counterfactual could have been raising spending such that ex-ante value function of firms stay the same, which would mean entry-exit incentives remain unchanged. In that setting, the equilibrium quality would rise, and we can compare the additional spending with the increase in quality. I choose to not do that because a lot of discussion in the US – specifically around hospices – are about how to reduce healthcare spending. This counterfactual suggests a way to do so without sacrificing quality.

life as a result. I also run a descriptive regression to show that visits-per-patient-day seems to have gone up after this policy was implemented, though the effect is weak (Table 39).[36] This suggests that a hybrid-scheme can boost visits when applied over the patient's entire length-of-stay, which would also minimize provider concerns about scrutiny or predicting the last week of life.

6.3 Information-based policy counterfactuals

Next, I study information-based policy tools. In 2017 CMS launched a website called "Hospice Compare" (now "Care Compare" with a hospice category). It reports a caregiver-survey summary and some measures of hospice quality using claims data from prior two years (Appendix M). However, the caregiver statistics appears to be missing for most hospices. It is also difficult to cleanly gauge the effect of this website.[37]

Since the website aims to aggregate quality-history information about hospices, and thus improve consumer choice based on quality-history, this suggests several research questions. First, what could be the effect of improving quality-history accessibility and salience through such websites on market outcomes? In particular, since consumers are now more likely to switch to better quality-history hospices, would it also affect the quality of service by the hospices themselves? Second, can such quality-history information interact with Medicare schemes to affect equilibrium quality?

These questions also have implications for releasing review websites for other healthcare providers, not just hospice care. Furthermore, if the market outcomes improve from policymaker's perspecive, then there should be further promotion of Hospice Compare, encouragement of consumer participation, and additional improvements such as better coverage of caregiver survey data.

This is also motivated by the prior literature on public report cards. Jin and Leslie (2003), Jin and Sorensen (2006), and Dafny and Dranove (2008) all find that introducing public report cards (containing information on last hygiene inspection, past consumer satisfaction rates, etc.) leads to some consumers switching to higher quality firms; Chartock (2025) finds that disclosing quality ratings leads to significantly higher demand at highly-rated doctors. Importantly, Jin and Leslie (2003) further show that it induces firms to raise quality.

To study this, I increase the strength of quality-history effects in consumer choice by raising

[36]The lack of strong evidence in my descriptive regression is unsurprising. First, anecdotally it appears the SIA has seen little uptake, due to worry about payor scrutiny and difficulty in predicting the week of death of a patient (Appendix G.4.2). Second, since it is a nationwide policy, I am estimating the effect off of time-series variation. As a result, and because it occurs near the end of my dataset, I do not model SIA in this paper.

[37]This is for several reasons. First, many of the main quality measures from claims-data were rolled out during 2022, and so the measures and market shares are affected by COVID. Second, sine this is a review website, this is a national rollout and so there is no clear control group. Third, the caregiver survey is missing for many hospices. Finally, the rollout of the quality measures was staggered – the caregiver survey rollout happened around 2017, while the claims-based quality measures began in 2022.

η_2. Recall that information on a_{jt} and Γ_{jt} are obtained from two separate sources: a_{jt} is obtained from visiting or contacting the hospice and its current patients, while Γ_{jt} is obtained from social workers, support groups, and the community. The "review website" thought experiment is that it makes information on Γ_{jt} more accessible and salient to consumers, thereby increasing the effect of Γ_{jt} in consumer choice. This is modeled by increasing η_2 in the demand model.

Strengthening quality-history effects: Within raising η_2 to simulate stronger quality-history effects, I conduct two sets of counterfactuals. The first set (Table 13) is for a market with three identical firms, while the second set (Table 14) is for a market with one high-cost and one low-cost hospice. The latter counterfactual is motivated by the fact that there is heterogeneity in per-visit costs within a market.

In both cases, I find that stronger quality-history effects lead to higher equilibrium quality. This is because the marginal gain from raising quality increases: an increase in quality leads to a larger increase in future sales, so it is optimal to incur the higher cost from higher quality.

Table 13: Impact of stronger quality-history effects with homogeneous firms.

η_2 scaled	Quality	Market Share	Ex-Ante Value Function (normalized)	Medicare Spending (normalized)
1.0	0.688	0.204	1.0	1.0
1.25	0.709	0.227	1.023	1.114
1.5	0.731	0.246	1.04	1.206
1.75	0.745	0.258	1.053	1.262
2.0	0.758	0.269	1.065	1.319

Notes: The column "η_2 scaled" represents the constant by which η_2 is scaled up to amplify quality-history effects.

Table 14: Quality-history effect and reallocation with heterogeneous firms.

η_2 scaled	Quality: Firm 1	Quality: Firm 2	Market Share: Firm 1	Market Share: Firm 2
1.0	0.666	0.773	0.217	0.392
2.0	0.731	0.8	0.241	0.563

Notes: The column "η_2 scaled" represents the constant by which η_2 is scaled up to amplify quality-history effects. The model is solved for a duopoly with one high-cost firm (Firm 1) and one low-cost firm (Firm 2).

In the homogeneous firm case, it is also interesting to consider the impact on market share and Medicare spending. As quality-history effects become stronger, Medicare spending increases even though per-diem Medicare rates stay the same. This is because of the market expansion effect

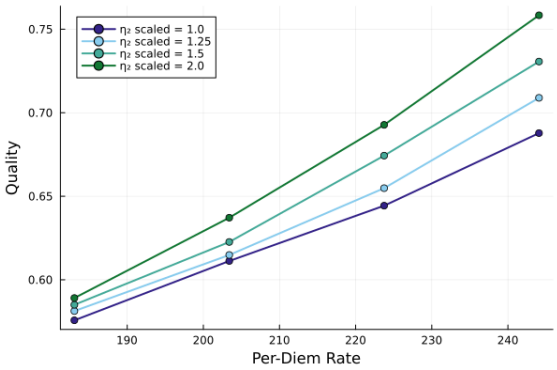
– as quality-history effects become stronger, more consumers switch from the outside option to hospices, leading to more hospice patients and greater Medicare spending. This is further amplified by hospices providing higher quality due to stronger quality-history effects. All this can be seen in the variable "market share", which shows how even among identical hospices the market shares of individual hospices increase. This is driven by market share taken from the outside option.

For the heterogeneous firm case, I consider a duopoly with one high-cost firm and one low-cost firm (Table 14). I find that market share increases significantly for the low-cost firm as quality-history effects become stronger, while the high-cost firm's market share increases only slightly. The latter happens because the high-cost firm draws some market share from the outside option, allowing it to overcome the business-stealing effects from the low-cost firm.

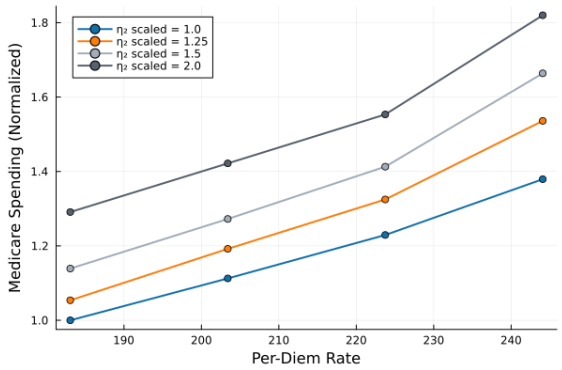
Interaction between reimbursement rates and quality-history effects I investigate the interaction between Medicare reimbursement rates and quality-history effects.

First, I look at how the quality response to higher Medicare reimbursement rates varies as quality-history effects become stronger (Figure 3 for triopoly, Figure 7 for duopoly). I find that the quality response to higher Medicare rates is stronger when quality-history effects are stronger.

Figure 3: Impact of larger quality-history effects and reimbursement rates.



(a) Quality vs per-diem rates by η_2 scaled



(b) Medicare spending vs per-diem rates by η_2 scaled

Notes: These graphs show how equilibrium quality and Medicare spending change as Medicare reimbursement rates increase, with different lines representing different levels of quality-history effect η_2. Here "η_2 scaled" represents the constant by which η_2 is scaled up to amplify quality-history effects.

Second, I study if Medicare reimbursement rates can be reduced without sacrificing quality by strengthening quality-history effects. I find that as quality-history effects become stronger, the per-diem rate can be reduced while maintaining the same equilibrium quality choice. However, Medicare spending increases as quality-history effects become stronger because of the market expansion effect – stronger quality-history effects lead to more consumers switching from the

Table 15: Trading off stronger quality-history effects with lower per-diem rates.

η_2 scaled	Per-Diem Rate	Per-Diem Rate (normalized)	Medicare Spending (normalized)
1.0	244.08	1.0	1.0
1.25	236.96	0.971	1.031
1.5	233.3	0.956	1.108
1.75	231.27	0.948	1.185

Notes: All counterfactuals maintain an equilibrium quality of 0.685. The column "η_2 scaled" represents the constant by which η_2 is scaled up to amplify quality-history effects.

outside option to hospices, even though equilibrium quality stays the same.

6.4 Discussion

Additional discussion on the hybrid reimbursement scheme: A possible issue with subsidizing visits is that it could encourage spurious visits or fraudulent reporting by hospices. However, I believe that is less of a concern for the following reasons:

1. As mentioned before, we can interpret the counterfactual as showing we should tie reimbursement to some measure of quality, not necessarily visits.

2. My estimates suggests that quality-history effects are salient in this industry. Thus, a hospice that makes spurious visits can worsen their quality-history and lose future consumers, which might rein in such issues.

3. Medicare already face issues like fraudulent reporting and upcoding in other settings. Therefore, to discourage such actions, Medicare can deploy the sort of documentation and monitoring that it already employs. For example, i) it can adopt finer contract structures (such as reimbursing hospices for up to 3 visits a week), ii) it can enforce stronger monitoring and auditing, for instance the patient's family may be asked to sign off after every visit.

Finally, another possible advantage of the hybrid reimbursement scheme is that it may reduce incentives for hospices to avoid patients with short lengths-of-stay who require many visits (Gruber et al. (2023)).

Medicare hospice spending versus total spending: Several counterfactuals show Medicare spending on hospices rises as we change policy levers. However, this does not necessarily mean that total Medicare/payor spending rises as well. For instance, patients may be switching out of expensive ineffective care in the outside option and into hospice care, which would raise Medicare's

hospice spending but lower total Medicare/payor spending. This is in contrast to other healthcare industries, such as the nursing home industry, where Hackmann (2019) shows people substitute from cheaper community-based care to more expensive nursing home care, raising overall Medicaid spending.

7 Conclusion

Using firm-level data from California, I study hospice quality choices, the role of quality-history in consumer demand, and policies that can improve quality while reducing Medicare spending. I develop and estimate a structural model of consumer demand and hospice quality choice, measuring quality as visits per patient per day and defining a hospice's quality-history as a function of its past quality choices. Consistently providing high-quality care therefore builds a stronger quality-history over time.

The estimates show that quality-history substantially influences consumer choice and is highly persistent. The estimated cost of an additional patient visit ranges from $153 to $256, depending on the hospice and county.

Counterfactual analyses show that stronger consumer responses to quality-history and higher Medicare reimbursement rates increase hospice quality. Quality-history effects also amplify the effect of reimbursement rates on quality. Comparing the current daily Medicare payment scheme with a hybrid scheme combining daily and per-visit payments, I find that the hybrid scheme can achieve the same level of hospice quality at lower Medicare spending.

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Online Appendix

A Additional data preparation for structural estimation

Censoring quality variable: To deal with high outlier values of the quality choice variable, I censor the visits-per-patient-day variable at the 90th percentile. That is, for quality values which are greater than the 90th percentile, I recode those values to be equal to the 90th percentile value.

Missing observations and fixing Firm IDs: I perform some additional data-cleaning before doing the analysis. First, for 24 hospices, it seems from looking at the data that a hospice forgot to respond in a certain year. This conjecture is due to the fact that the hospice was operating and serving a large number of patients in both the past few and next few years, but suddenly had its observation missing in one year. These non-responses are scattered across several years, with most in 2003 and only one after 2010. For these instances, I copy over the hospice's previous year's sales and characteristics into the missing year. This comprises 24 imputed data points in my dataset of approximately 2000 observations.

The data provider HCAI assigns an ID to each hospice, called the OSHPD ID. For 3 OSHPD IDs, they disappear from the data for 3-4 years before returning. I classify these as new entries by different entities.

Finally, for a few hospices, their OSHPD IDs change despite everything else about them staying the same (their name, location, etc.). For these hospices, I harmonize each to have the same OSHPD ID throughout.

Imputing quality-history of older hospices: For hospices which enter after 2002 (earliest period in my dataset), I can construct their exact quality-history (Γ_{jt}) because I set their quality-history during entry to be 0. However, a number of hospices are present from before the year 2002, and for them, I do not observe their pre-2002 quality choices.

While this does not affect my demand estimation (observe that the Berry (1994) allows me to difference out log of outside option share to account for rivals' characteristics), it will prevent me from conducting my supply side estimation. This is because a hospice's quality choice in each observed state is influenced by all its rivals' quality-histories.

As a result, for the year 2002, I impute the quality-history Γ_{jt} of hospices arriving before 2002 using a flexible functional form. First, for all hospices that arrived *after* 2002, I regress their

quality-history on a large number of observed covariates. I use these regression coefficients to predict the quality-history – for the year 2002 – of all hospices that arrived before 2002. Once I know their quality-history coming into year 2002, I can use all their successive quality choices to update their quality-history in future periods.

B Empirical policy function

I regress the quality measure visits-per-patient-day on a variety of covariates to see how quality choice is affected by state variables. This sheds light on how hospices react to quality-history levels and product characteristics of themselves and rivals. I interpret this regression as reflecting the policy function used by hospices when making quality decisions.[38]

I use the demand estimates from the previous section to construct quality-history Γ_{jt} of each firm in every market-year, the attractiveness of their product attributes ($X'_{jt}\beta_d$), and their hospice-specific demand unobservables (ξ_{jt}). In a dynamic game, these values of a hospice and its rivals would enter the policy function.

To account for competitive effects from rivals, I include the following state variables for a given hospice: number of rivals, number of rivals squared, difference between mean Γ_{jt} of rivals and own, difference between mean $X'_{jt}\beta_d$ of rivals and own, and difference between mean ξ_{jt} of rivals and own.

I approximate the empirical policy function using two different methods: a 2SLS regression (with a variety of fixed effects for robustness; see below) and a flexible regression using second-order polynomials. Below I explain the exact implementation of both methods.

Both specifications for estimating policy function yield very similar results for the slope of the cost function. I generally find that number of rivals is the key driver of the cost estimation results, and that is preserved with both the 2SLS and other flexible functions I have tried.

B.1 Policy function using 2SLS

When recovering the empirical visit-choice policy function from the data, I try to be cognizant of the issues raised in Berry and Compiani (2023) and Berry and Compiani (2021). Briefly, a regression of visit choice on dynamic states such as quality-history can suffer from endogeneity problem in the presence of serially correlated unobservables.[39] To deal with such issues, I take the

[38] Note that for-profit status changes over time for same hospice, allowing me to identify the for-profit coefficient even with hospice fixed effects.

[39] For instance, if the unobserved shocks are serially correlated and also affect the visit choice, then the dynamic state would be influenced by past shocks; past shocks are correlated with current shocks; and so the current period dynamic state is correlated with the current shock.

following steps. First, I include fixed effects (at the county-level or firm-level) in my regression. Second, following the suggestions from Berry and Compiani (2023), I use instruments to correct the endogeneity problem. I use lagged $X'_{jt}\beta_d$ and lagged ξ_{jt} to instrument for the dynamic state (difference between mean Γ_{jt} of rivals and own). These instruments are valid because current period's quality choice is affected by *current* $X'_{jt}\beta_d$ and ξ_{jt}, while the last period's quality choice (which features in Γ_{jt}) is affected by last period's $X'_{jt}\beta_d$ and ξ_{jt}. (Note that current period's $X'_{jt}\beta_d$ and ξ_{jt} are included in the regression as well.) Lagged $X'_{jt}\beta_d$ and lagged ξ_{jt} also affect rivals' last period quality choice, which features in rivals' Γ_{jt}.

Berry and Compiani (2023) go further and suggest using nonparametric IV methods such as GIV, with their argument being that the policy function of the firm may be highly nonlinear. While I run a linear regression to simplify estimation, I conduct the following check. In my counterfactual, I solve for the policy function for heterogeneous hospices. Then, I check the sign on the policy function for different values of the state variables (own quality-history, rival quality-history, own and rival $X'_{jt}\beta_d$, own and rival ξ_{jt}). I find that the signs on the solved policy function match the signs on the regression, i.e. they agree on which direction each state variable moves the quality choice. Thus, while the *exact* policy function may be nonlinear, I argue that this regression represents a meaningful approximation of it.

The resulting 2SLS regression has signs on the state variables that match the general shape of the policy functions I observe in my counterfactuals.[40] Furthermore, I check the 2SLS first-stage regression of dynamic states on the exogenous variables and instruments, and find them to have signs predicted by economic theory; see Appendix B.1.1 for results and discussion. The results of the regression are given in Table 16. To summarize, hospices choose higher quality when facing more rivals, and when facing rivals with better quality-history, $X'_{jt}\beta_d$, and ξ_{jt}. They choose lower quality when they themselves have higher quality-history, $X'_{jt}\beta_d$, and ξ_{jt}.

Note that while I use this empirical policy function for the estimation, I do not use it for my counterfactuals; instead, I will numerically solve for the optimal policy function.

The first stage of Bajari et al. (2007) requires an estimate of the policy function from the data. To this end, I use a variant of the empirical policy function in Table 16. Recall that I discretize visit choice into 6 tiers; as a result, I use the empirical policy function and random draws from the residual distribution to make predictions of quality choice given values of state variables, then round it to the nearest quality tier.

[40] In additional counterfactuals (not reported in the main text, available upon request), I solve for the policy function for heterogeneous hospices. I find that, holding other state variables constant, a hospice chooses higher quality if its rivals have high quality-history/$X'_{jt}\beta_d/\xi_{jt}$, and lower quality if the hospice itself has high quality-history/$X'_{jt}\beta_d/\xi_{jt}$.

Table 16: Estimates from 2SLS regression of visit-per-patient-day on covariates.

	Visits-per-patient-day		
	(1)	(2)	(3)
Count of rivals	0.120***	0.076**	0.060**
	(0.021)	(0.026)	(0.020)
Count of rivals squared	-0.003***	-0.002*	-0.002*
	(0.001)	(0.001)	(0.001)
For-profit	0.034	-0.115	-0.147*
	(0.035)	(0.085)	(0.071)
Medicare rate $\times\overline{LOS}$	5.68×10^{-6}	-4.01×10^{-6}	-3.06×10^{-7}
	(3.57×10^{-6})	(7.05×10^{-6})	(5.60×10^{-6})
Mean $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$	0.149**	0.390*	0.300*
	(0.045)	(0.155)	(0.118)
Mean ξ_{jt} of rivals - own ξ_{jt}	0.702***	0.487**	0.395***
	(0.102)	(0.150)	(0.114)
Mean Γ_{jt} of rivals - own Γ_{jt}	0.337***	0.448**	0.349**
	(0.072)	(0.173)	(0.131)
log(reported per-visit cost)			-0.210***
			(0.027)
Firm Fixed Effects	Yes		
County Fixed Effects		Yes	Yes
Obs.	1,608	1,634	1,634

Notes: Estimates of fixed effects suppressed. The endogenous variable in the regression is the dynamic state "Mean Γ_{jt} of rivals - own Γ_{jt}". The instruments used are ξ_{jt-1} and $X'_{jt-1}\beta_d$. The per-visit cost and intercept cost used in this regression are those reported by the hospices in the cost data. $\overline{LOS}$ is set to 49.16 days.

B.1.1 First stage of 2SLS

In this subsection I dig into the 2SLS regression reported in Table 16. The endogenous variable in that regression was the dynamic state "Mean Γ_{jt} of rivals - own Γ_{jt}". The instruments used were ξ_{jt-1} and $X'_{jt-1}\beta_d$.

Below I disentangle the first stage regression for the 2SLS. Fundamentally, I want to make sure that the instruments are moving the dynamic state in accordance with economic theory. Briefly, a higher ξ_j or $X_j\beta_d$ should cause hospice j to choose lower quality and therefore get lower quality-history. On the other hand the rivals would choose higher quality and therefore improve their quality-history. These are insights I get from conducting counterfactuals involving heterogeneous hospices.

I therefore run the first-stage IV regression separately on two variables: Mean Γ_{jt} of rivals and own Γ_{jt}. The tables below show the regression of each variable on the exogenous variables and

instruments. They confirm that the instruments move the dynamic states in the direction I expect.

Table 17: Estimates from first stage of 2SLS regression.

	Mean Γ_{jt} of rivals - own Γ_{jt}		
	(1)	(2)	(3)
ξ_{jt} (lagged)	0.233***	0.197**	0.209**
	(0.042)	(0.067)	(0.067)
$X'_{jt}\beta_d$ (lagged)	0.302	0.275	0.293
	(0.267)	(0.306)	(0.305)
Firm Fixed Effects	Yes		
County Fixed Effects		Yes	Yes
Obs.	1,608	1,634	1,634

Notes: Estimates of fixed effects and other covariates suppressed.

Table 18: Estimates from first stage of 2SLS regression.

	Mean rival Γ_{jt}		
	(1)	(2)	(3)
ξ_{jt} (lagged)	0.103***	0.065*	0.066*
	(0.031)	(0.032)	(0.032)
$X'_{jt}\beta_d$ (lagged)	0.320	-0.350*	-0.349*
	(0.196)	(0.146)	(0.146)
Firm Fixed Effects	Yes		
County Fixed Effects		Yes	Yes
Obs.	1,608	1,634	1,634

Notes: Estimates of fixed effects and other covariates suppressed.

B.2 Flexible policy function

As an alternative, I use a flexible function of the same state variables to approximate the policy function. I regress quality on county fixed effects and second-degree polynomials of most of the covariates. There are advantages to this as well. It better captures any nonlinearities that the policy function may exhibit and therefore avoids functional form misspecification issues. The disadvantage is I cannot control for serially correlated unobservables by using instrumental variables.

Table 19: Estimates from first stage of 2SLS regression.

	Γ_{jt}		
	(1)	(2)	(3)
ξ_{jt} (lagged)	-0.130***	-0.131*	-0.143**
	(0.035)	(0.055)	(0.055)
$X'_{jt}\beta_d$ (lagged)	0.017	-0.625*	-0.642*
	(0.218)	(0.251)	(0.250)
Firm Fixed Effects	Yes		
County Fixed Effects		Yes	Yes
Obs.	1,608	1,634	1,634

Notes: Estimates of fixed effects and other covariates suppressed.

Table 20: Estimates from flexible regression.

	Visits-per-patient-day
Count of rivals	0.018**
	(0.006)
Count of rivals squared	-4.5 $\times 10^{-4}$
	(2.3 $\times 10^{-4}$)
County Fixed Effects	Yes
Controls	Yes
Obs.	1,797

Notes: Estimates of county fixed effects suppressed. Controls are for-profit status and second-degree polynomials of the following variables: Medicare-rates, Mean $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$, Mean ξ_{jt} of rivals - own ξ_{jt}, and Mean Γ_{jt} of rivals - own Γ_{jt}.

B.3 Entry policy function

Table 21: Ordered logit of entry count on market characteristics, used as entry transition function.

	Entry count
Firm count	-0.300
	(0.117)
Market size	0.001
	(3.461×10^{-4})
Medicare price	-3.846×10^{-4}
	(2.423×10^{-4})
County FE	Yes

Notes: For each market-year, I calculate the total number of entries ("entry count"). Then I run an ordered logit of entry counts on market characteristics (market fixed effects, market size, Medicare price). I restrict the ordered logit to market-years with 3 or less entries; greater than 3 entries are very rare and give unintuitive results. Estimates of county fixed effects and cutoffs suppressed.

B.4 Results with firm-fixed effects in policy function regression

With regards to the empirical policy function that uses firm fixed effects, I do not make use of the estimated firm-specific fixed effects for making quality predictions during the simulation. Most firms are observed for a few periods, and so the fixed effect estimates are likely to be biased. Also, as alluded to earlier, the fixed effects are identified off of an assumption of linear policy function, which is unlikely to hold in practice. The advantage of having fixed effects is that it allows me to control for unobserved heterogeneity across firms that could be causing endogeneity problems. During cost estimation, I find that the cost function estimates are similar with or without firm fixed effects.

Table 22: Results of supply estimation with firm fixed effect in empirical policy function.

	(1)
Per-visit cost	229.482
	(8.716)
Low-cost county	-77.635
	(13.333)
θ_e	22.276
	(4.728)

Notes: Specification for empirical policy functions uses firm fixed effects, instruments, reported per-visit costs, and other covariates. See Table 16 for the resulting empirical policy function with firm fixed effects. All empirical poicy functions use covariates that convey information about level of competition, own quality-history, and rival quality-history. See Appendix B for full details. Per-visit costs and logit scaling parameter θ_e (in $\times 10^5$) are estimated using the methods of Bajari et al. (2007) and Pakes et al. (2007). Standard errors (in parentheses) are calculated via bootstrapping at the county-year level. 200 bootstrap runs were used.

C Robustness checks for demand estimation

I conduct additional robustness checks for demand estimation. First, I reestimate demand without imposing the AR(1) structure on ξ_{jt} and continue finding evidence of quality-history; see Table 25. Second, instead of BLP-IVs, I redo my demand estimation with Gandhi-Houde IVs (Gandhi and Houde (2019)) and obtain similar results. I also use two additional instruments – i) reported per-visit cost for each hospice from the cost data, and ii) Medicare prices. The reported per-visit cost acts as a cost shifter. Medicare prices impact quality choice (see counterfactuals) while being orthogonal to firm-specific demand shocks since they are set at the national level and vary by county based on Medicare cost indices. Thus they vary quality by making patients more/less profitable to serve or by indicating potentially higher costs of providing care. See Berry and Haile (2014) for the importance of cost-shifters in random coefficients discrete choice model estimation.

I also repeat my demand estimation with county-dummy-by-fuel-price IVs. All specifications yield similar results; see tables below for these and other robustness checks. In particular, I use different subsets of the aforementioned instruments to ensure my results aren't driven by any specific instrument. Finally, a previous version of the paper used "visits-per-patient" as the quality choice variable and obtained similar results.

C.1 With persistence of ξ

I estimate the following system of equations using 2-step GMM.

$$ln(s_{jt}) - ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} + \eta_1 a_{jt} + \eta_2 \Gamma_{jt}$$

$$\xi_{jt} = \rho \xi_{jt-1} + \varepsilon_{jt}$$

Here τ denotes depreciation rate of quality-history, ρ denotes persistence of unobserved hospice-specific demand shocks, σ_n is the nest parameter. Γ_{jt} is calculated as $\Gamma_{jt} = a_{jt-1} + (1-\tau)a_{jt-2} + (1-\tau)^2 a_{jt-3} + \ldots = \sum_{i=1}^{age_{jt}} (1-\tau)^{i-1} a_{jt-i}$.

For robustness, I re-estimate this model using different sets of instruments. I continue to find evidence of quality-history and its persistence.

Also note that the absence of quality-history persistence would mean $\tau = 1$; therefore the t-stat for τ is $\frac{\hat{\tau}-1}{s.e.(\tau)}$.

Table 23: Results of demand estimation with BLP-IVs.

	(1)	(2)	(3)
τ	0.194	0.247	0.121
	(0.096)	(0.097)	(0.104)
η_1	0.155	0.505	0.481
	(0.181)	(0.249)	(0.213)
η_2	0.204	0.276	0.204
	(0.079)	(0.111)	(0.097)
σ_n	0.798	0.661	0.668
	(0.054)	(0.107)	(0.108)
ρ	0.893	0.680	0.589
	(0.071)	(0.115)	(0.134)
For-profit	0.121	-0.039	-0.002
	(0.079)	(0.106)	(0.099)
Hospice inpatient unit	0.149	0.156	0.171
	(0.113)	(0.124)	(0.140)
Pediatric program	0.101	0.192	0.168
	(0.086)	(0.120)	(0.128)
Bereavement services	-0.087	-0.069	-0.075
	(0.028)	(0.021)	(0.023)
Day care for adults	0.069	0.092	0.043
	(0.111)	(0.123)	(0.134)
Agency type: free-standing	0.013	-0.050	-0.071
	(0.113)	(0.116)	(0.099)
Agency type: home health based	0.044	-0.043	-0.043
	(0.127)	(0.112)	(0.112)
Agency type: hospital-based	0.095	0.020	-0.002
	(0.129)	(0.152)	(0.138)

Notes: (1): BLP-IV with county-by-fuel IVs and reported per-visit cost and inflation-adjusted Medicare prices; (2): BLP-IV with reported per-visit cost and inflation-adjusted Medicare prices; (3): BLP-IV with reported per-visit cost. All specifications include lagged rival count and lagged reported per visit cost as IVs. The demand model is estimated using GMM. Bootstrapped standard errors are in parentheses. Estimates of county fixed effects are suppressed.

Table 24: Results of demand estimation with GHIVs.

	(1)	(2)	(3)
τ	0.162	0.231	0.123
	(0.099)	(0.095)	(0.099)
η_1	0.083	0.440	0.533
	(0.122)	(0.251)	(0.256)
η_2	0.178	0.278	0.243
	(0.073)	(0.096)	(0.100)
σ_n	0.842	0.645	0.585
	(0.047)	(0.082)	(0.078)
ρ	0.914	0.573	0.265
	(0.059)	(0.277)	(0.309)
For-profit	0.171	-0.044	-0.074
	(0.062)	(0.146)	(0.149)
Hospice inpatient unit	0.148	0.161	0.187
	(0.110)	(0.114)	(0.129)
Pediatric program	0.067	0.204	0.215
	(0.075)	(0.072)	(0.055)
Bereavement services	-0.094	-0.067	-0.070
	(0.033)	(0.029)	(0.028)
Day care for adults	0.068	0.056	-0.017
	(0.118)	(0.120)	(0.118)
Agency type: free-standing	0.031	-0.067	-0.130
	(0.105)	(0.150)	(0.156)
Agency type: home health based	0.075	-0.064	-0.111
	(0.126)	(0.163)	(0.161)
Agency type: hospital-based	0.123	-0.012	-0.086
	(0.118)	(0.199)	(0.203)

Notes: In what follows, GHIV stands for Gandhi-Houde IVs (Gandhi and Houde (2019)). (1): GHIV with county-fuel IVs and reported per-visit and inflation-adjusted Medicare prices; (2): GHIV with reported per-visit and inflation-adjusted Medicare prices, (3):GHIV with reported per-visit. All specifications include lagged rival count and lagged reported per visit cost as IVs. The demand model is estimated using GMM. Bootstrapped standard errors are in parentheses. Estimates of county fixed effects are suppressed.

C.2 Without imposing persistence of ξ

I estimate the following system of equations using GMM.

$$ln(s_{jt}) - ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} + \eta_1 a_{jt} + \eta_2 \Gamma_{jt}$$

I no longer impose the autoregressive structure on ξ_{jt}. This means I now interact the instruments with the ξ_{jt} term directly, rather than the innovation of ξ_{jt} as in the previous subsection.

Here τ denotes depreciation rate of quality-history, σ_n is the nest parameter.

For robustness, I re-estimate this model using different sets of instruments. I continue to find evidence of quality-history.

Also note that the absence of quality-history persistence would mean τ = 1; therefore the t-stat for τ is $\frac{\hat{\tau}-1}{s.e.(\tau)}$.

Table 25: Results of demand estimation without imposing persistence of ξ.

	(1)	(2)	(3)	(4)
τ	0.272	0.010	0.290	0.138
	(0.072)	(0.048)	(0.047)	(0.046)
η_1	0.903	0.470	0.814	0.641
	(0.259)	(0.224)	(0.224)	(0.272)
η_2	0.335	0.157	0.373	0.271
	(0.121)	(0.053)	(0.082)	(0.049)
σ_n	0.543	0.649	0.531	0.557
	(0.121)	(0.093)	(0.066)	(0.062)
For-profit	-0.184	0.020	-0.190	-0.112
	(0.100)	(0.111)	(0.121)	(0.147)
Hospice inpatient unit	0.163	0.192	0.168	0.192
	(0.132)	(0.154)	(0.126)	(0.156)
Pediatric program	0.262	0.156	0.271	0.226
	(0.129)	(0.122)	(0.078)	(0.110)
Bereavement services	-0.057	-0.077	-0.057	-0.071
	(0.016)	(0.022)	(0.025)	(0.033)
Day care for adults	0.131	-0.027	0.096	-0.012
	(0.129)	(0.166)	(0.130)	(0.153)
Agency type: free-standing	-0.106	-0.113	-0.119	-0.148
	(0.120)	(0.122)	(0.137)	(0.146)
Agency type: home health based	-0.107	-0.066	-0.125	-0.123
	(0.109)	(0.158)	(0.154)	(0.179)
Agency type: hospital-based	-0.031	-0.057	-0.056	-0.095
	(0.195)	(0.182)	(0.219)	(0.213)

Notes: (1): BLP-IVs and reported per-visit cost and Medicare rates; (2): BLP-IVs; (3): GH-IVs and reported per-visit cost and Medicare rates;(4): GH-IVs. All specifications use lagged count of rivals and lagged reported per-visit-cost as IVs. The demand model is estimated using GMM. Bootstrapped standard errors are in parentheses. Estimates of county fixed effects are suppressed.

C.3 Setting $\eta_1 = 0$

Table 26: Results of demand estimation with BLP-IVs and GH-IVs, imposing $\eta_1 = 0$.

	(1)	(2)
τ	0.159	0.166
	(0.100)	(0.112)
η_2	0.200	0.205
	(0.108)	(0.119)
σ_n	0.782	0.779
	(0.082)	(0.056)
ρ	0.798	0.770
	(0.096)	(0.174)
For-profit	0.130	0.126
	(0.077)	(0.105)
Hospice inpatient unit	0.158	0.158
	(0.116)	(0.110)
Pediatric program	0.113	0.117
	(0.096)	(0.061)
Bereavement services	-0.084	-0.084
	(0.024)	(0.035)
Day care for adults	0.001	0.001
	(0.078)	(0.084)
Agency type: free-standing	-0.010	-0.011
	(0.121)	(0.147)
Agency type: home health based	0.013	0.010
	(0.133)	(0.175)
Agency type: hospital-based	0.043	0.041
	(0.152)	(0.195)

Notes: (1): BLP-IV with reported per-visit cost and inflation-adjusted Medicare prices; (2): GH-IV with reported per-visit cost and inflation-adjusted Medicare prices. All specifications use lagged count of rivals and lagged reported per-visit-cost as IVs. The demand model is estimated using GMM. Estimates of county fixed effects are suppressed.

D Alternative quality-history definitions

Table 27: Results of demand estimation with $\psi_{jt} = \eta a_{jt} + (1-\tau)\psi_{jt-1}$.

	(1)	(2)	(3)	(4)
τ	0.268	0.228	0.236	0.137
	(0.089)	(0.096)	(0.070)	(0.073)
η	0.401	0.353	0.372	0.307
	(0.148)	(0.140)	(0.101)	(0.102)
σ_n	0.666	0.670	0.650	0.588
	(0.105)	(0.108)	(0.081)	(0.082)
ρ	0.705	0.688	0.590	0.311
	(0.094)	(0.098)	(0.220)	(0.251)
For-profit	-0.029	-0.013	-0.034	-0.053
	(0.100)	(0.106)	(0.140)	(0.150)
Hospice inpatient unit	0.156	0.161	0.162	0.191
	(0.124)	(0.139)	(0.112)	(0.128)
Pediatric program	0.195	0.187	0.203	0.221
	(0.119)	(0.126)	(0.077)	(0.056)
Bereavement services	-0.068	-0.071	-0.067	-0.069
	(0.021)	(0.025)	(0.030)	(0.029)
Day care for adults	0.072	0.045	0.041	-0.079
	(0.100)	(0.123)	(0.097)	(0.096)
Agency type: free-standing	-0.049	-0.056	-0.066	-0.138
	(0.119)	(0.105)	(0.160)	(0.171)
Agency type: home health based	-0.048	-0.050	-0.066	-0.129
	(0.117)	(0.120)	(0.175)	(0.175)
Agency type: hospital-based	0.009	-0.001	-0.018	-0.120
	(0.166)	(0.141)	(0.222)	(0.229)

Notes: This defines $\psi_{jt} = \eta a_{jt} + (1-\tau)\psi_{jt-1} = \eta \sum_{i=0}^{age_{jt}} (1-\tau)^i a_{jt-i}$. (1): BLP-IV with reported per-visit cost and inflation-adjusted Medicare prices; (2): BLP-IV with reported per-visit cost; (3): GH-IV with reported per-visit cost and inflation-adjusted Medicare prices (4): GH-IV with reported per-visit cost. All specifications use lagged count of rivals and lagged reported per-visit-cost as IVs. The demand model is estimated using GMM. Estimates of county fixed effects are suppressed.

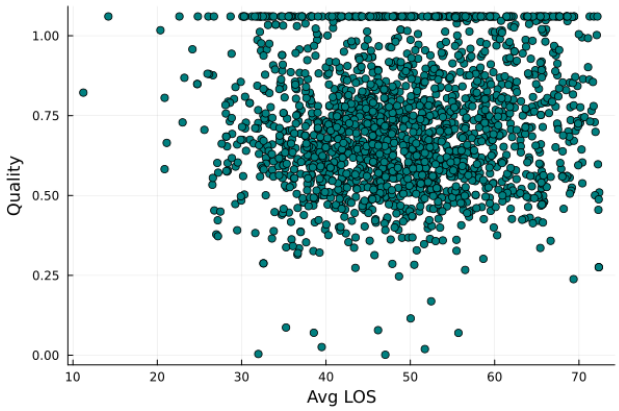
E Additional descriptive statistics and regression results

Table 28: Regressing market share on age of hospices.

	log(market share)
Age	0.344***
	(0.030)
Age (squared)	-0.016***
	(0.002)
County Fixed Effects	Yes
Obs.	1,002

Notes: Standard errors clustered at county-level in parentheses.

Figure 4: Scatterplot of average length-of-stay and quality.



Notes: "Avg LOS" stands for average length-of-stay.

Table 29: Regressing quality on average lengths-of-stay.

	Visits-per-patient-day		
	(1)	(2)	(3)
Count of rivals	0.014***	0.009**	0.009**
	(0.002)	(0.003)	(0.003)
Avg LOS	-0.0003	-0.001	-0.005
	(0.001)	(0.001)	(0.004)
Avg LOS squared			4.2×10^{-5}
			(3.9×10^{-5})
County Fixed Effects	Yes	Yes	Yes
Controls		Yes	Yes
Obs.	1,956	1,956	1,956

Notes: This regression drops observations of average lengths-of-stay above the 95th percentile (approximately 71+ days) to prevent outliers affecting the results. Standard errors clustered at county-level in parentheses. "Avg LOS" stands for average length-of-stay. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators.

Table 30: Regression of market share on quality and lagged quality, using IVs.

	log(market share)		
	(1)	(2)	(3)
Count of rivals	-0.054***	-0.057**	-0.059**
	(0.013)	(0.017)	(0.018)
Visits-per-patient-day	1.091**	1.184**	
	(0.381)	(0.403)	
Sum of past 3 visits	0.476*	0.568*	
	(0.230)	(0.224)	
Log of visits-per-patient-day			0.724*
			(0.274)
Log of sum of past 3 visits			1.319*
			(0.503)
County Fixed Effects	Yes	Yes	Yes
Year Fixed Effects		Yes	Yes
Controls	Yes	Yes	Yes
Obs.	1,403	1,403	1,403

Notes: This regression instruments for current quality and stock of past qualities with current and lagged reported per-visit cost (as cost shifters). The notation Visits-per-patient-day$_{t-n}$ denotes the visits-per-patient-day chosen n years prior. Sum of past 3 quality choices = $\sum_{i=1}^{3}$ Visits-per-patient-day$_{t-i}$. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators. Standard errors clustered at county-level in parentheses.

E.1 Length of stay

Length of stay	% of patients
0-7 days	30.89
8-30 days	29.38
31-90 days	21.7
91-179 days	9.52
180+ days	8.51

Table 31: Distribution of patient length-of-stay.

In my dataset, I observe the number of patients in a hospice who stayed for i) 0-7 days, ii) 8-30 days, iii) 31-90 days, iv) 91-179 days, v) 180+ days. Table 31 uses this data to construct the total number of patients in my dataset within each LOS bracket. Over 80% of the patients stay for less than 90 days.

Table 32 shows the distribution of people within each LOS-bracket by hospice-year. For nearly all hospices, most of their patients stay for less than 90 days.

	10%	25%	50%	75%	90%
0-7 days	19.76	24.66	30.3	36.31	41.81
8-30 days	21.93	25.05	29.29	33.01	36.33
31-90 days	16.15	18.9	21.7	25.17	29.44
91-179 days	5.45	7.47	9.41	11.43	13.92
180 days	2.24	4.48	7.2	10.73	14.52

Table 32: Distribution of shares of lengths-of-stay of patients. Each observation is at the hospice-year level.

E.2 Disease category

Diagnosis	% of patients
Cancer	39.6
Heart	10.27
Dementia	13.57
Lung	6.39
Kidney	2.49
Brain stroke	4.47

Table 33: Percentage of hospice patients by diagnosis.

The dataset classifies the patients at a hospice by broad catergories of diagnosis. This allows me to see which diagnosis are more prevalent and whether hospices specialize in certain diagnosis over others. Table 33 shows that most hospice patients suffer from cancer; the second and third largest diagnosis categories are heart disease and dementia.

To see if hospices are specializing by diagnosis, or whether patients select into hospices by their diagnosis, Table 34 shows the distribution of patients of each diagnosis by hospice-year. This shows that most of a hospice's patients are suffering from cancer, heart disease or dementia, and there is no clear sign that a hospice gets most of its patients from one disease category alone.

	10%	25%	50%	75%	90%
Cancer	24.23	31.25	39.63	49.42	60.0
Non cancer	40.0	50.52	60.36	68.75	75.73
Heart disease	3.91	6.67	9.81	13.29	17.82
Dementia	3.33	6.5	11.72	18.03	25.0
Lung disease	2.61	4.27	6.18	8.46	11.12
Kidney	0.0	1.34	2.32	3.49	5.21
Liver	0.0	0.98	1.85	2.91	4.3
Brain stroke	0.0	1.53	3.48	5.89	8.96
HIV	0.0	0.0	0.0	0.13	0.46
Coma	0.0	0.0	0.0	0.0	0.35
Diabetes	0.0	0.0	0.0	0.0	0.44
ALS	0.0	0.0	0.26	0.61	1.03

Table 34: Distribution of shares of patients at a hospice per disease category. Each observation is at the hospice-year level.

E.3 Care type

	10%	25%	50%	75%	90%
Routine care	98.72	99.46	99.8	99.94	100.0
Inpatient care	0.0	0.0	0.04	0.23	0.69
Respite care	0.0	0.0	0.06	0.16	0.36
Continuous care	0.0	0.0	0.0	0.01	0.08

Table 35: Distribution of days of service provided for each hospice-year per care-type.

Medicare reimburses hospices with rates based on whether it provided routine care, inpatient care, respite care, and continuous care. Routine care has the lowest Medicare reimbursement rate, while other care types are much higher (see Figure 8).

The dataset divides the total days of care provided by a hospice into days providing each type of care. I calculate the fraction of care days in my dataset for each type of care. Table 35 shows that nearly all of a hospice's care days are for routine care.

	10%	25%	50%	75%	90%
Registered nurses	8.82	11.42	14.44	18.21	22.48
Licensed vocational nurses	0.0	0.0	0.64	4.43	8.9
Homemakers	5.39	7.69	11.24	16.85	24.21
Physicians	0.0	0.0	0.01	0.25	0.77
Social service workers	1.74	2.59	3.55	4.73	6.05
Chaplains	0.27	1.02	1.84	2.87	4.08

Table 36: Distribution of visits by staff type for each hospice-year. The unit is visit per patient.

	Min	10%	25%	50%	75%	90%	Max
Share	0.0	0.0	0.0	0.00546448	0.0136986	0.0291228	0.590909

Table 37: Distribution of share of patients discharged to other hospices for each hospice-year.

E.4 Market definition and selection

E.4.1 Defining markets to be at the county level

While I define my market to be at the level of a county, some papers on healthcare providers define markets at a larger level. Furthermore, some hospices advertise as being available for service in multiple counties at once.

In my setting, I can rule out larger market definitions using my dataset. For each hospice, the dataset includes measures of the fraction of patients that arrive from each county. This allows me to isolate the fraction of a hospice's patients coming from the "home county" of the hospice (i.e. the county where the hospice is located) versus the fraction that is coming from an "away county".

Variable	5%	10%	25%	50%	75%	90%
Home County Share	0.39	0.54	0.74	0.91	0.99	1.0

Table 38: Distribution of share of patients coming from home county. Observations are at the hospice-year level.

As Table 38 make clear, the vast majority of a hospice's patients arrive from the county where it is located. As a result, I define the relevant market to be at the county level.

For the hospices which are below the reported 5th percentile, I see which counties they are located in. This is to check if there is one specific county where nearly all hospice patients are from outside that county. That turns out not to be the case – the hospices with home county share below the 5th percentile are scattered across over 10 counties.

E.4.2 Market selection

I focus on the following 39 counties in California: Alameda, Contra Costa, Amador, Butte, San Joaquin, El Dorado, Fresno, Humboldt, Del Norte, Imperial, Kern, Kings, Lake, Madera, Marin, Mariposa, Mendocino, Merced, Monterey, Napa, Nevada, Placer, Sacramento, San Francisco, San Luis Obispo, San Mateo, Santa Barbara, Santa Clara, Santa Cruz, Shasta, Siskiyou, Solano, Sonoma, Stanislaus, Sutter, Tehama, Tulare, Tuolumne, Yolo.

I drop counties which saw very high number of hospices by the end of the dataset's coverage (2018). The counties which are dropped for this purpose are (with their number of hospices in 2018 in parentheses) are: Los Angeles (377), Orange (63), Riverside (34), San Bernardino (92), San Diego (31), and Ventura (40).

The highest number of hospices for a county in my dataset is 23 (Alameda). Therefore, I am implictly dropping counties which had more than 23 hospices by the end of the dataset's coverage.

How representative is having a sample limited to 23 or fewer hospices? For that, I look at the number of hospices by 3-digit zip code in the US for the year 2025.[41] The 90th percentile of that distribution has 17 hospices in a 3-digit zip code.

E.4.3 Discussion on sample selection

I now expand on why I leave out large markets with many hospices from my sample:

1. Large markets have different competition and dynamics compared to smaller markets. This could mean the effect of quality-history is harder to estimate in such settings.

   For example, in large markets, search frictions are likely to be strong. As an example, Ventura has 40 hospices, Orange has 60+, and San Bernardino even more. It is highly unlikely that consumers (or the intermediaries helping them make decisions) are aware of all these hospices when making decisions. These search frictions and consideration choice set issues may therefore introduce noise in consumer choices and dampen quality-history competition. If so, nailing down the demand coefficients on quality-history is significantly difficult and plagued by noise.

   Furthermore, in the presence of such search frictions, hospices may also compete very differently. For example, some hospices might chance having lower quality and still get many patients. Or, hospices might spend more time networking with nursing homes/physicians rather than focus on quality to get patient referrals. This means that the policy function of quality in such settings may be quite different than in smaller markets.

2. Another reason for leaving out large counties is the nonstationarity that comes with it. Many of these counties are growing quickly, with the number of hospices rising rapidly over my dataset.[42] For instance, between 2002 to 2017, San Bernardino went from 6 to 84 hospices, Orange went from 11 to 54 hospices, etc. This is very different from the more steady dynamics observed in smaller markets. This suggests that the competition and dynamics in these markets are very different and so shouldn't be modeled together. My stationary dynamic model is more appropriate for the smaller markets and is not suited to handling the nonstationary dynamics in these large markets.

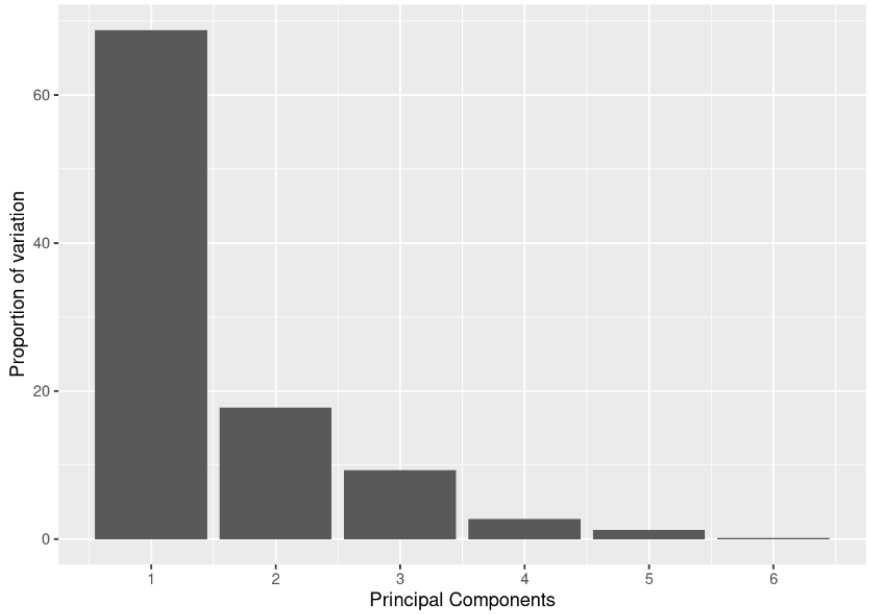
3. These larger markets are also often with much larger populations. It is possible that competition is more localized; i.e., the market definition may be smaller than at the county level.

[41] Data obtained from the following link: https://data.cms.gov/provider-characteristics/hospitals-and-other-facilities/hospice-enrollments/data

[42] I shed some light on this in a current working paper, Alam (2025).

How representative is having a sample limited to 23 or fewer hospices? For that, I look at the number of hospices by 3-digit zip code in the US for the year 2025.[43] The 90th percentile of that distribution has 17 hospices in a 3-digit zip code.

E.5 Principal components analysis of visits by staff type



	PC1	PC2	PC3	PC4	PC5	PC6
rnpat	0.4223153	0.7939506	-0.3565230	0.2495343	0.0433376	0.0061776
physnpat	0.0151012	-0.0068927	-0.0221099	-0.0032503	-0.0416916	-0.9987426
socsvcspat	0.1317572	0.1580743	-0.1338480	-0.9503556	0.1912078	-0.0010246
lvnpat	0.1834381	-0.5006202	-0.8411163	0.0710564	0.0519645	0.0224485
homemkrpat	0.8730719	-0.3053850	0.3769924	0.0254161	0.0411040	0.0051643
chaplainpat	0.0903201	0.0269610	-0.0699097	-0.1698567	-0.9774627	0.0440833

Figure 5: Principal components analysis of visits by staff type. The staff types are (in order of presentation) Registered Nurse, Physician, Social service worker, Licensed Vocational nurse, Home-maker, and Chaplains. The PCA analysis shows that the first Principal Component explains most of the variation in visits by a hospice.

[43] https://data.cms.gov/provider-characteristics/hospitals-and-other-facilities/hospice-enrollments/data

F Descriptive regressions for counterfactuals

Table 39: Regression of market-level quality on covariates.

	log(mean quality)
Count of firms	0.057***
	(0.012)
Count of firms squared	-0.002***
	(0.000)
Mean reported per-visit cost	-2.59 $\times 10^{-5}$*
	(1.22 $\times 10^{-5}$)
log(Medicare rate)	-0.419
	(0.254)
year	4.747
	(2.894)
$year^2$	-0.001
	(0.001)
year == 2016	0.059
	(0.055)
year > 2016	0.123*
	(0.062)
County Fixed Effects	Yes
Obs.	635

Notes: This regression is at the market-year level. The dependent variable and reported per-visit cost are averaged at the market-year level. Polynomials of year are included to control for time trends. The dummy for 2016 is included to allow for delayed adoption or awareness of SIA.

Table 40: Regression of market-level quality on Medicare rates and other covariates.

	log(mean quality)
(Intercept)	-0.457
	(0.281)
log(count of firms)	0.063***
	(0.011)
log(mean reported per-visit cost)	-0.149***
	(0.034)
log(medicare rate)	0.135***
	(0.040)
Obs.	635

Notes: All variables are averaged at the market-year level.

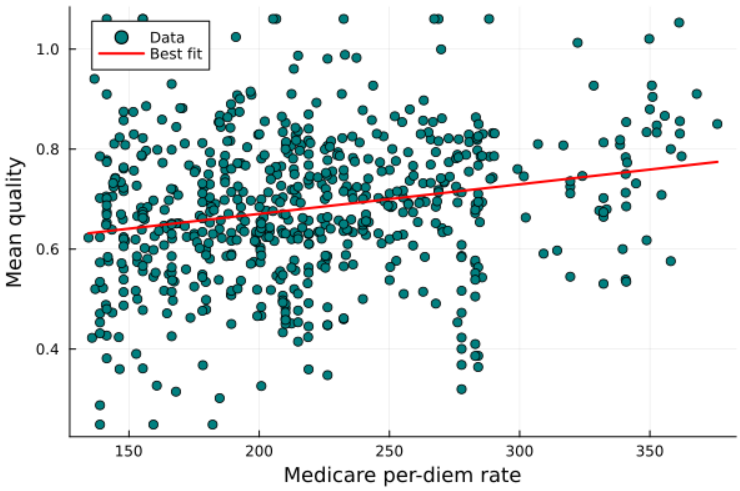


Figure 6: Scatterplot of market-level quality against Medicare rates. Each point is a market-year. The line is the best linear fit.

G Additional discussions

G.1 Unobserved demand shocks

The AR(1) process governing ξ_{jt} captures persistent unobservables in demand. One example could be that a hospice is well-connected with physicians and social workers in some hospitals and so get repeated referrals despite not choosing higher quality - this will manifest as the hospice having high ξ_{jt} for several periods. Another example could be a hospice that has suffered a recent scandal such as elder abuse. Despite choosing high visits-per-patient-day the negative publicity may last for several periods - this will manifest itself as the hospice having low ξ_{jt} for several periods.

While the above can be achieved without explicitly modeling ξ_{jt} as persistent, the AR(1) structure presents an additional advantage. It allows for the possibility that a hospice with high ξ might face less entry because it is more difficult to compete against. Without it, I would have to assume that level of competition – which is my main source of instruments – in the market is orthogonal to the ξ of all incumbents. But that might not be true: if the incumbent has persistently high (or low) ξ, then that could affect entry/exit decisions and hence the level of competition in the market. By allowing for persistence, the claim is that level of competition is orthogonal to some residual term after accounting for persistence. In the demand estimation this means I build moments around ε_{jt} instead of ξ_{jt}; this follows Grennan (2013) and Hodgson (2023).

I now discuss the distinction I make between quality-history (ψ_{jt}) and unobserved demand shock (ξ_{jt}) of a hospice. Quality-history is built up through a hospice's efforts over time, and my measure of visits-per-patient-day attempts to proxy for this effort. The unobserved demand shock is not interpreted as capturing unobserved quality or effort dimensions, but rather captures non-quality dimensions such as connectedness with physicians, scandals, factors outside the hospice's control, etc. As a result, ξ_{jt} is not interpreted as contributing to a hospice's quality-history.

That said, to the extent that there are dimensions of quality not proxied by my quality measure that help build quality-history, those will likely get subsumed in the AR(1) process of ξ_{jt}. While this is hopefully less of an issue in hospice care, where the type of care provided is mostly low skill, it is nevertheless something I cannot conclusively rule out. If better data were available on the various dimensions of quality, I could include those to create a more comprehensive quality measure for my structural model.

Finally, I reestimate demand without imposing the AR(1) structure on ξ_{jt}, thus building moments around ξ_{jt} instead of ε_{jt}, and continue finding evidence of quality-history (Table 25).

G.2 Discussion on capacity constraints

Table 41: Regression of total visits by a hospice in a year on covariates.

	Total visits		log(total visits)	
	(1)	(2)	(3)	(4)
Patients	53.992***	44.326***		
	(3.766)	(7.030)		
Count of rivals	2186.478***	1929.452***		
	(489.214)	(503.053)		
Count of rivals squared	-62.318***	-52.588**		
	(15.967)	(17.719)		
log(patients)			1.104***	1.059***
			(0.016)	(0.038)
log(total firms)			0.252**	0.193**
			(0.072)	(0.062)
County Fixed Effects	Yes		Yes	
Year Fixed Effects	Yes	Yes	Yes	Yes
Firm Fixed Effects		Yes		Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,060	2,034	2,060	2,034

Notes: Standard errors clustered at county-level in parentheses. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators.

There may be some concern about whether capacity constraints are present in this industry. That is, each hospice really has a fixed number of staff over time, and those staff get spread out over however many patients the hospice has in any given year.

I claim that capacity constraints are not a concern in my setting. This is for the following reasons:

1. **Market share is positively correlated with visits-per-patient-day:** If capacity constraint was present, it would create negative correlation between visits-per-patient-day and market share. The reason is that as a hospice gets more patients, it can make fewer visits to each of them due to the same number of staff getting spread out more thinly across a larger patient pool. However, the pattern I see in my data is the opposite. I find that market share is *positively* correlated with visits-per-patient-day; see Table 5. This suggests that hospices with more patients also make more visits per patient, which is consistent with the idea that they are not capacity constrained.

2. **Firms increase the total visits they make when facing greater competition:** After controlling for total patients, I see total number of visits by a hospice increases as there are more competitors. The results are reported in Table 41. This suggests that, holding fixed the patient volume, a hospice makes more visits when facing more rivals in the market. This goes against the idea that hospices have a fixed number of staff as competitors enter and that these staff are distributed over whatever number of patients the hospice has. Instead, it suggests that hospices respond to entry by making more visits in total.

3. **Institutional details suggest capacity constraint is unlikely to be an issue:** Hospices are not really limited by physical space since they generally serve patients at patient residences. From my conversations with a hospice admin and from online readings, it seems that hospices hire nurses living around the area where their patients reside. The hospice nurses often drive directly from their home to various patient residences. Therefore, there isn't a capacity constraint with hospices like there is with nursing homes or hospitals.

G.3 Heuristic discussion on identification

Intuitively, what is the variation in data that lets me pin down key demand parameters and cost parameters?

In a simplified sense, demand parameters are identified by regressing current market share on a function of current and past quality choices. The nonlinear regression tells me how much impact past choices have on current share, as well as how important the total quality-history stock is for explaining current market share. The nonlinear regression is a more rigorous version of the linear regression shown in Table 5. The quality-history stock ψ_{jt} varies separately from ξ_{jt} because it subsumes current and past quality choices by the hospice.

The cost parameters are identified through a revealed preference argument. Let us think about the cost of increasing quality from 0.44 to 0.56. Accounting for the demand system and rivals, an increase in quality will lead to higher market share. So the main reason I would observe a firm choosing 0.44 instead of 0.56 is if the increase in total revenue is more than offset by the corresponding increase in cost.

G.4 Additional discussion on institutional details and modeling decisions

G.4.1 Advertising

I allow for unobserved demand shocks in my demand model, and advertising gets picked up there. Furthermore, my understanding is that advertising was pretty low in the hospice industry during my data period; in fact, hospices have generally suffered from lack of awareness. I don't model advertising as a strategic choice variable in my supply estimation.

G.4.2 Additional dimension of Medicare hospice reimbursement

There are additional dimensions of the Medicare reimbursement scheme that I do not study in this paper. First, the total revenue paid by Medicare to a hospice is subject to a cap that rises with the number of patients enrolled in the hospice. For a detailed study, see Gruber et al. (2023). Second, from 2016, Medicare began paying hospices an additional amount for visiting a patient in the last week of her life (called the "service intensity add-on"). Third, Medicare also lowers the per-diem rate for routine care if a patient is enrolled for more than 60 days to discourage hospices from enrolling non-terminal patients. Interestingly, it seems from news reports in 2019 that the service intensity add-on had seen limited uptake from hospices due to "... logistics and staff shortages to providers' anxieties about potential payor scrutiny and the inherent difficulty of predicting when a

patient will expire."; see https://hospicenews.com/2019/05/20/medicare-service-intensity-add-on-underused-by-hospices/.

H Model estimation and solution

H.1 Demand model

For the demand estimation, I limit my dataset to hospices that enter after 2002 (the earliest period in my data). For these hospices, I see the quality choices over their lifetime, and so can construct Γ_{jt} accurately for my demand estimation. Please see Appendix A for a more thorough discussion.

For demand estimation, I estimate the following set of equations:

$$ln(s_{jt}) - ln(s_{0m(j)t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} + \eta_1 a_{jt} + \eta_2 \Gamma_{jt}$$

$$\xi_{jt} = \rho \xi_{jt-1} + \varepsilon_{jt}$$

This model is estimated via nonlinear GMM. Let the matrix of instruments be Z. I set the weighting matrix to be the inverse of $Z'Z$.

GMM requires estimating a large number of parameters, including county fixed effects and coefficients on hospice characteristics. A trick to easing computational burden is to note that only η_1, η_2, τ and ρ enter nonlinearly, so I can concentrate out all the linear parameters and minimize over these nonlinear parameters. To concentrate out the linear parameters, I can use the assumption that X_{jt} and ξ_{jt} are orthogonal to each other, so market fixed effects and coefficients on hospice characteristics can be estimated via OLS regression.

H.2 Supply estimation

H.2.1 How market share is calculated from demand estimates

To calculate profit, I need to calculate market shares of all firms for a given state $\mathbf{x}_m$. This is done using the analytical formulae for nested logit (see Berry (1994)). Let

$$\delta_{jt} = \alpha_m + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \psi_{jt} + \xi_{jt}$$

where σ_n is the nest parameter that captures correlated preferences within the nest. Then the market share can be written as:

$$s_j(\delta, \sigma_n) = \bar{s}_{j/g}(s, \sigma_n)\bar{d}_g(\delta, \sigma_n) = \frac{e^{\delta/(1-\sigma_n)}}{D_g^{\sigma_n}\left[\sum_g D_g^{(1-\sigma_n)}\right]}$$

where

$$D_g \equiv \sum_{j \in G_g} e^{\delta_j/(1-\sigma_n)}$$

H.2.2 Equilibrium concepts

Markov Perfect Equilibrium: Markov Perfect Equilibrium is a refinement of Nash Equilibrium that is a type of Subgame Perfect Equilibrium. It says that a player's strategy depends only on "payoff-relevant" state variables rather than on potentially the whole history of play. Intuitively, it limits the number of variables that a player looks at before taking an action. This introduces simplicity as well as ruling out a vast number of multiple equilibria. See Ericson and Pakes (1995) for a detailed discussion.

Symmetric and anonymous policy functions: I also impose the policy functions to be symmetric and anonymous. The symmetry assumption imposes that all firms have the same policy function. The anonymity assumption imposes that a firm does not condition its choice on the identity of its rivals.

H.2.3 Forward simulation and optimization

A brief overview of the algorithm is as follows. Fix a guess of structural parameters θ. For every observed state $\mathbf{x}_t$ and hospice j (i.e. each observation in my hospice-level panel data), I estimate the choice-specific value function for each quality level. For quality choice a_j by hospice j, a single simulated path is constructed as follows:

1. The first-stage policy function is used to predict current quality choices by all firms, conditional on the simulated market state in that period.

2. Market share of each hospice is realized. Each incumbent receives a payoff.

3. The entry transition function is used to predict the number of potential entrants who apply for entry. These entries are implemented in the next period.

4. The empirical exit probabilities are used to predict whether any incumbent exits.

5. Using the state variable transition functions from first-stage, all state variables update. This includes unobserved demand shocks, market size, and Medicare rates. Γ_{jt} is updated according to the transition equation $\Gamma_{t+1} = a_t + (1-\tau)\Gamma_t$.

6. The game moves to the next period, and I repeat this algorithm.

The above simulation is run for many periods until the discounted profit is driven close to zero; in my estimation I set this to be 80 periods. The simulations use Halton sequences for making all random draws. Summing together the discounted per-period payoffs gives me the total payoff from hospice j choosing quality a_j for a single simulated path. I repeat this simulation 200 times and average over the total payoffs. This gives me the choice-specific value function of hospice j choosing quality a_j in state $\mathbf{x}_t$, $\hat{v}_j\left(a_j, \mathbf{x}_t, \sigma^*; \theta\right)$. In short, I average over the sum of discounted payoffs for each simulation run, and this gives me the choice-specific value function for that action-state combination.

The choice-specific value functions are constructed for every quality $a \in \mathscr{A}$. These are then used to construct model-predicted choice probabilities $\Psi_j\left(a_j, \mathbf{x}_t, \sigma^*; \theta\right)$ from the structural model using Equation (11). The model-predicted choice probabilities are interpreted as follows – given state $\mathbf{x}_t$, how likely is it that hospice j would choose action a_j, conditional on our guess of structural parameters θ.

Next, I describe how these model-predicted choice probabilities are used to estimate the hospice cost function. The intuition is that I search for values of parameters θ that minimize the distance between model-predicted choice probabilities and observed choices.

The dynamic oligopoly model is estimated using GMM. The GMM residual term for observation n is:

$$\Xi_n(\theta) = a_n^{data} - \sum_{a_n \in \mathscr{A}} a_n \hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$$

where $\hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$ is the predicted choice probability for action a_n, a_n^{data} is the observed quality choice for observation n, θ is the guess of structural parameters, and $\mathbf{x}_n^{data}$ is the market state for observation n. The resulting GMM optimization problem is:

$$\min_{\theta} \left[\frac{1}{N}\sum_n Z_n' \Xi_n(\theta)\right]' \hat{W} \left[\frac{1}{N}\sum_n Z_n' \Xi_n(\theta)\right]$$

This is solved with GMM. The instruments used in GMM are the county indicators and the variables in the first-stage empirical policy functions (with the exception of the firm fixed effects), giving us the moment conditions $E[Z'\Xi_n(\theta)]$.

Let the matrix of instruments be Z. I set the weighting matrix to be the inverse of $Z'Z$.

A way I ease computational burden is by using the "linear-in-parameters" method outlined in Bajari et al. (2007) and Ryan (2012). Similar to their setup, the hospice payoff functions can be split up in such a way that I do not need to redo the forward simulation for each guess of the cost parameters. Instead, I can perform the forward simulation just once and then use the trial cost

parameters to rescale the choice-specific value functions and then evaluate the objective function. Please see Bajari et al. (2007) and Ryan (2012) for a careful explanation.

H.3 Bootstrapping standard errors

Following Wang (2022), I sample blocks of county-year observations from my data with replacement until I match the total number of county-year observations in my dataset. Then I re-estimate the model on each bootstrap sample. I use the sample variance over all the resulting bootstrap estimates to calculate the standard errors for my cost parameters.

H.4 Counterfactual model solution via policy iteration

H.4.1 Discretizing Γ_{jt}

To simplify computation, I discretize quality-history Γ_{jt}, and impose a stochastic evolution process to minimize the effects of any such discretization. This closely follows Benkard (2004).

The stochastic evolution works as follows. Fix a firm, an industry state, and a chosen action a_t. Let Γ_t be the quality-history predicted by the quality-history transition equation $\Gamma_{t+1} = a_t + (1-\tau)\Gamma_t$. Let Γ_d be the largest quality-history less than Γ_{t+1} and Γ_u be the lowest quality-history greater than Γ_{t+1}. In the counterfactual, next period's quality-history Γ^*_{t+1} then follows the stochastic process below:

$$\Gamma^*_{t+1} = \begin{cases} \Gamma_u & \text{w.p.} \quad \frac{\Gamma_{t+1}-\Gamma_d}{\Gamma_u-\Gamma_d} \\ \Gamma_d & \text{w.p.} \quad 1-\frac{\Gamma_{t+1}-\Gamma_d}{\Gamma_u-\Gamma_d} \end{cases} \tag{15}$$

H.4.2 Policy iteration

To conduct counterfactuals I solve the model by policy iteration method. This requires solving a coupled fixed-point problem; more specifically, I need to calculate $\Psi_j\left(a_j, \mathbf{x}_m, \boldsymbol{\sigma}^*\right)$, $\hat{v}(a_j, \mathbf{x})$ and $V(\mathbf{x})$ such that Equations (16), (17), and (18) simultaneously hold:

$$\Psi_j\left(a_j, \mathbf{x}_m, \boldsymbol{\sigma}^*\right) = \frac{\exp\left\{\hat{v}_j\left(a_j, \mathbf{x}_m, \boldsymbol{\sigma}^*\right)/\theta_e\right\}}{\sum_{a\in\mathscr{A}}\exp\left\{\hat{v}_j\left(a, \mathbf{x}_m, \boldsymbol{\sigma}^*\right)/\theta_e\right\}} \tag{16}$$

$$\hat{v}_j(a_j, \mathbf{x}_m, \boldsymbol{\sigma}^*) = \sum_{\mathbf{a}_{-j}\in\mathscr{A}_{-j}} \left\{ \left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) + \beta V(\mathbf{x}'_m, \boldsymbol{\sigma}^*)\right] F(\mathbf{x}'_m|\mathbf{x}_m, a_j, \mathbf{a}_{-j}) \prod_{n\neq j} \Psi_n(\mathbf{a}_{-j}[n], \mathbf{x}_m, \boldsymbol{\sigma}^*) \right\} \tag{17}$$

where $\Psi_n(\cdot, \mathbf{x}_m, \sigma^*)$ is the CCP of firm n, $\mathscr{A}_{-j}$ is the set of all possible rival action profile vectors, and $\mathbf{a}_{-j}[n]$ is the action by firm n.[44]

$$V_j(\mathbf{x}_m, \sigma^*) = \theta_e \left[0.577216 + \ln \Big(\sum_{a_j \in \mathscr{A}} e^{\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*)/\theta_e} \Big) \right] \tag{18}$$

I solve the model by iterating between the three equations until convergence. Policy iteration proceeds as follows. I iterate between Equations (17) and (18) several times, then iterate between Equations (16) and (17) several times. I repeat this until successive CCPs and ex-ante value functions are within some pre-determined tolerance. I use the Anderson algorithm to accelerate convergence.

H.5 Absence of multiple equilibria

Estimation and solution of dynamic games are sometimes hampered by presence of multiple equilibria. While I do not prove uniqueness of equilibria, this appears to not be an issue in my setting:

1. Some multiple equilibria are also ruled out under the assumptions made in Appendix H.2.2, which are standard in the dynamic game estimation literature.

2. To see if my specific model of dynamic games in my setting have multiple equilibria, I explored extensively with the dynamic game solution via policy iteration to see if the algorithm ever converges to different equilibria. I did this for heterogeneous firms, varying numbers of firms, different starting points for the policy iteration algorithm, etc. In all the cases I tried, the algorithm converged to the same equilibrium.

3. Multiple equilibria might have been an issue if firms chose both prices and quality; in such cases, one firm might choose to be low-price low-quality-history, while another might choose to be high-price high-quality-history. This would lead to multiple equilibria. However, in the hospice industry, firms choose only quality; the price is set by Medicare.

That said, I cannot rule out the possibility of multiple equilibria completely, since I have not proved uniqueness of equilibria mathematically.

[44] Prior to knowing the choice-specific structural errors, $\hat{v}(a_j, \mathbf{x}, \sigma^*)$ is hospice j's expected discounted sum of payoffs from choosing action a_j when facing state variables $\mathbf{x}$. Forming this expectation requires integrating over several random variables. The last term shows that this expression requires integrating over rival actions, which is done using the CCP at equilibrium strategies. Intuitively, since this is a simultaneous-move game, when hospice j chooses an action this period it does not know its rivals actions, and so has to make predictions using the equilibrium strategies. The middle term shows that conditional on own choice and rival choices, the firm also needs to integrate over all possible values of state variables. The first term is then the current period payoff and continuation value conditional on state variables and rival choices.

I Interpreting the structural model and counterfactuals

I.1 Interpreting entry-exit incentives from ex-ante value functions

I report the ex-ante value functions for my counterfactuals to show how entry-exit incentives change. The overall intuition is that as market conditions improve, potential entrants will choose to enter; as market conditions worsen, incumbents will choose to exit.

I now give two examples to elaborate.

1. In a sequential entry game of homogeneous firms with no private shocks (e.g. Igami (2018)), the equilibrium number of firms n^* is given by:

$$V(n^*, s_0) \geq \kappa_{entry}$$

where $V(\cdot,\cdot)$ is the value function, s_0 is the starting state, and κ_{entry} is the entry cost. We can see here how a policy change that uniformly raises $V(\cdot,\cdot)$ will raise n^*, and the opposite will lower n^*.

2. In dynamic games studying exit, the exit decision compares the value function from staying on with the scrap value, i.e. $V(x_m) \leq \phi_{scrap}$. If this inequality holds, then subject to logit shocks, the firm is likely to exit. Any policy that lowers $V(\cdot)$ therfore makes exit more likely.

I.2 Interpreting the profit function

Let the profit from patient i be

$$\Pi_i = P_m^{MCAR} LOS_i - \gamma_0 - \gamma_1 a_j LOS_i$$

This is a random variable, becaues LOS_i varies by patient i, and is unknown to the hospice when it chooses quality a_j or even during patient enrollment.

The expected profit per patient integrates over the above expression with respect to the distribution of LOS:

$$\mathbb{E}_{LOS}[\Pi_i] = P_m^{MCAR} \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}$$

where $\mathbb{E}_{LOS}[LOS_i] \equiv \overline{LOS}$ to match the notation in the main text.

Using this notation, the profit function used in the paper is therefore:

$$\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \mathbb{E}_{LOS}[\Pi_i]$$

$$\implies \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} P_m^{MCAR}\, \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}$$

$$\implies \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) \left[P_m^{MCAR}\, \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}\right]$$

which is the profit function reported in the main text.

Note that this is different from the profit function being the sum of *realized* profits per patient:

$$\tilde{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + \left\{ \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}$$

The expectation of this profit function lines up with the sum of expected profits:

$$\mathbb{E}_{LOS}[-\phi_j + \left\{ \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}] = -\phi_j + M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) \left[P_m^{MCAR}\, \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}\right]$$

However, it also exhibits variance.

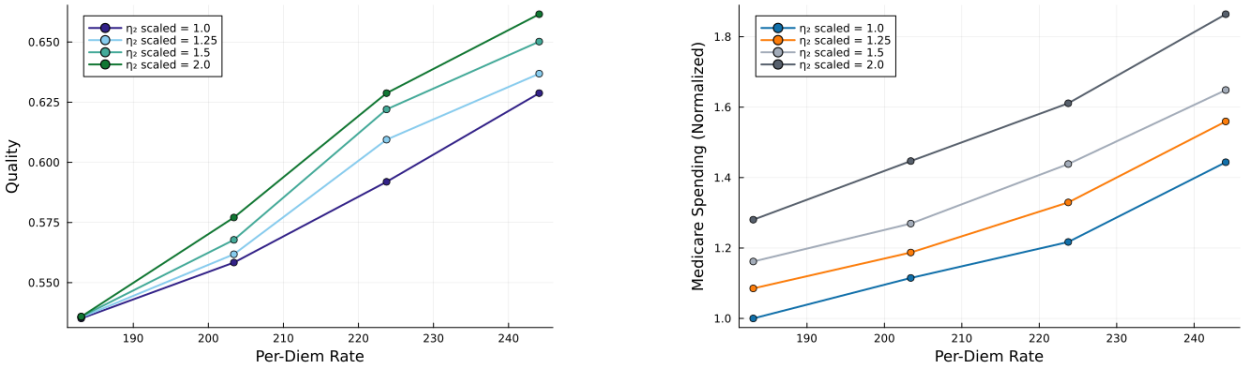
$$\mathbb{V}_{LOS}[-\phi_j + \left\{ \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}] \neq 0$$

while the sum of expected profits does not

$$\mathbb{V}_{LOS}[-\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \mathbb{E}_{LOS}[\Pi_i]] = 0$$

J Additional results on counterfactuals

Figure 7: Quality and Medicare spending variations in a duopoly



(a) Quality vs price by η_2 scaled

(b) Medicare spending vs price by η_2 scaled

Notes: These graphs show how equilibrium quality and Medicare spending change as Medicare reimbursement rates increase, with different lines representing different levels of quality-history effect η_2. Here "η_2 scaled" represents the constant by which η_2 is scaled up to amplify quality-history effects.

Table 42: Reimbursement schemes that lead to 0.687 visits-per-patient-day in a triopoly.

Per-Diem Rate	**Per-day rate**	**Ex-Ante Value Function (normalized)**	**Medicare Spending (normalized)**
244.08	0.0	1.0	1.0
211.33	30.0	0.912	0.948046
167.6	70.0	0.795	0.887329
123.46	110.0	0.677	0.825406

Notes: The rates are in US dollars. The total Medicare spending associated with each scheme is normalized with respect to the spending of the per-day scheme. The mean ex-ante value function is normalized with respect to the ex-ante value function of the per-day scheme. As we move down rows, equilibrium quality stays the same yet Medicare spending declines.

K Quotes on hospice's quality-history affecting consumer choice

1. Some websites which detail my claim on how hospices are chosen can be found in the following:

   - https://www.vitas.com/hospice-and-palliative-care-basics/when-is-it-time-for-hospice/how-to-choose-a-hospice-provider
   - https://www.cancer.org/treatment/end-of-life-care/hospice-care/how-to-find.html
   - https://www.health.harvard.edu/staying-healthy/choosing-hospice

- https://hospicefoundation.org/End-of-Life-Support-and-Resources/Coping-with-Terminal-Illness/How-to-Choose
- https://rainbowhospice.org/hospice-care/choosing-hospice/

2. Some quotes on how quality-history should be considered when choosing a hospice are as follows.

   - AmericanHospice.org: What do others say about this hospice? Get references both from people you know and from people in the field – e.g., local hospitals, nursing homes, clinicians. Ask anyone that you have connections to if they have had experience with the hospice and what their impressions are. Geriatric care managers can be a particularly good resource, as they often make referrals to hospices and hear from families about the care that was provided... How long has the hospice been in operation? If it has been around for a while, that's an indication of stability."
   - HospiceFoundation.org: "Seek professional opinions. Ask clinicians, professional caregivers at nursing homes, geriatric care managers, or end-of-life doulas about their experience with a hospice. Talk to friends, family, and neighbors who have used hospice services and get their opinions about the experience with a provider."
   - Vitas.com: "Evaluate the hospice provider's history and reputation before you decide. How long has it been in business? ... What do other patients or families say about their experiences?"
   - Caringinfo.org: "Most hospice programs use family satisfaction surveys to obtain feedback about their services so they can make improvements. Ask the hospice to share a summary of their family satisfaction scores for the last several months with you. You can also ask to see their latest state or Medicare inspection report to see if there are care provision problems. Finally, you could ask to see the hospice provider's list of complaints from the past 12 months."
   - Medicare provides a checklist of "Suggested Questions To Ask When Choosing A Hospice", and some of these questions refer to quality history: "How long has the hospice been serving patients?", "Can we speak with other caregivers to learn of their experience with the hospice?".

3. My claim is that visits-per-patient-day is a good proxy for the effort exerted by the hospice, since visits are usually the most costly part of care. Here is a quote from a New York Times article (https://www.nytimes.com/2023/06/10/health/hospice-profits.html) from a former hospice admin:

"In the nearly 20 years that Megan Stainer worked in nursing homes in and around Detroit, she could almost always tell which patients near death were receiving care from nonprofit hospice organizations and which from for-profit hospices."There were really stark differences," said Ms. Stainer, 45, a licensed practical nurse. Looking at their medical charts, "the nonprofit patients always had the most visits: nurses, chaplains, social workers. The nonprofit hospices responded quickly when the nursing home staff requested supplies and equipment."

K.1 Example of a hospice reimbursement scheme

LOCATION BY COUNTIES	ROUTINE HOME CARE DAYS 1-60 Revenue Code 0650	ROUTINE HOME CARE DAYS 61+ Revenue Code 0659	SERVICE INTENSITY ADD-ON (HOURLY) Revenue Code 0552	CONTINUOUS CARE (HOURLY) Revenue Code 0652	INPATIENT RESPITE CARE Revenue Code 0655	GENERAL INPATIENT CARE Revenue Code 0656
NATIONAL RATES	**$ 196.50**	**$ 154.41**	**$ 41.57**	**$ 41.57**	**$ 185.27**	**$ 758.07**
RURAL AREAS	$ 237.03	$ 186.26	$ 50.15	$ 50.15	$ 215.38	$ 903.74
ALAMEDA & CONTRA COSTA	$ 294.39	$ 231.33	$ 62.28	$ 62.28	$ 257.98	$ 1,109.87
BUTTE	$ 214.74	$ 168.74	$ 45.43	$ 45.43	$ 198.82	$ 823.63
FRESNO	$ 208.62	$ 163.94	$ 44.14	$ 44.14	$ 194.28	$ 801.64
IMPERIAL	$ 180.42	$ 141.77	$ 38.17	$ 38.17	$ 173.33	$ 700.28
KERN	$ 223.41	$ 175.56	$ 47.26	$ 47.26	$ 205.26	$ 854.78
KINGS	$ 206.61	$ 162.36	$ 43.71	$ 43.71	$ 192.78	$ 794.41
LOS ANGELES	$ 234.05	$ 183.92	$ 49.52	$ 49.52	$ 213.16	$ 893.02

Figure 8: Snapshot of Medicare reimbursement scheme in 2018. The reimbursement rate varies by county based on its Medicare wage index, and varies by the type of care provided.

L Hospice cost data

Per-period fixed cost:

- Sums: General service cost centers, Other Hospice Service Costs, Bereavement program costs, Other program costs, Writeoffs.
- Subtracts: Revenue from managed care, Revenue from private insurance, Revenue from self-pay, Revenue from other payers, Grants, Donations/Contributions, Unrelated Business Income, Total other operating revenue.

Figure 9: List of variables in the hospice cost data.

DETAIL OF OPERATING EXPENSES

(do not input "$" signs, commas, or decimals; round up to whole dollar)

Use data from Medicare Cost Report where applicable.

Line No.		Total (1)
	General Service Cost Centers	
30	Administrative and General	
	Inpatient Care Service	
31	Inpatient – General Care	
32	Inpatient – Respite Care	
	Program Supervision	
35	Hospice Program / Team Supervision (Non-Visit Wages)	
	Visiting Services	
36	Physician Services	
37	Nursing Care	
38	Rehabilitation Services (PT, OT, Speech)	
39	Medical Social Services – Direct	
40	Spiritual Counseling	
41	Dietary Counseling	
42	Counseling – Other	
43	Home Health Aides and Homemakers	
44	Other Visiting Services	
	Hospice Service Cost Centers	
45	Drugs, Biologicals, and Infusion	
46	Durable Medical Equipment / Oxygen	
47	Patient Transportation	
48	Imaging, Lab, and Diagnostics	
49	Medical Supplies	
50	Outpatient Services (including ER Dept.)	
51	Radiation Therapy	
52	Chemotherapy	
53	Other Hospice Service Costs	
	Other Hospice Costs	
54	Bereavement Program Costs	
55	Volunteer Program Costs	
56	Fundraising Costs	
	Other Costs	
57	Other Program Costs*	
59	Total Operating Expenses	

* Program costs including community education and outreach program costs.

Non-visit cost:

- Sums: Inpatient care service expense (both general and respite care), Radiation therapy cost, Chemotherapy cost, Program supervision cost, Volunteer program costs.
- This is divided by total number of patients in that year to get the non-visit cost per patient ("intercept cost").

Visit cost:

- Drugs/biologicals/infusion cost, Durable medical equipment/oxygen cost, Patient transportation cost, Imaging/lab/diagnostics cost, Medical supplies cost, Outpatient services cost, Visiting services cost (all staff types).
- This is divided by total number of visits in that year to get the cost of an additional visit ("reported per-visit cost").

Table 43: Regression of log of per-period fixed cost, log of intercept cost, and per-visit cost (from cost data) on county dummies.

	(1)	(2)	(3)
Alameda	13.920***	5.262***	111.949
	(0.111)	(0.157)	(102.670)
Amador	12.306***	6.163***	132.778
	(0.516)	(0.732)	(298.813)
Butte	13.109***	5.609***	118.234
	(0.214)	(0.303)	(167.659)
Contra Costa	13.718***	5.460***	141.194
	(0.133)	(0.188)	(122.592)
Del Norte	10.812***	5.493***	200.617
	(1.264)	(1.792)	(871.181)
El Dorado	13.447***	5.049***	128.679
	(0.298)	(0.422)	(275.492)
Fresno	13.043***	6.028***	114.276
	(0.140)	(0.198)	(129.868)
Humboldt	13.900***	5.157***	119.419
	(0.307)	(0.435)	(298.813)
Imperial	12.814***	7.155***	62.458
	(0.478)	(0.677)	(341.705)
Kern	12.943***	5.717***	370.076***
	(0.132)	(0.187)	(120.234)
Kings	13.280***	5.861***	316.966
	(0.351)	(0.497)	(298.813)
Lake	12.594***	5.407***	110.496
	(0.316)	(0.448)	(308.009)
Madera	9.335***	4.375**	132.881
	(1.264)	(1.792)	(550.983)
Marin	13.708***	6.830***	130.809
	(0.283)	(0.401)	(251.488)
Mariposa	11.003***	6.630***	115.431
	(0.381)	(0.540)	(275.492)
Mendocino	12.982***	6.192***	151.994
	(0.316)	(0.448)	(298.813)
Merced	12.737***	5.164***	100.129
	(0.243)	(0.345)	(202.545)
Monterey	13.732***	6.021***	177.939
	(0.202)	(0.287)	(177.829)
Napa	13.257***	5.133***	116.208
	(0.316)	(0.448)	(298.813)
Nevada	12.922***	6.513***	186.773
	(0.248)	(0.351)	(211.292)
Placer	13.732***	6.263***	160.865
	(0.184)	(0.261)	(177.829)
Sacramento	13.731***	5.434***	128.329
	(0.117)	(0.166)	(107.643)
San Francisco	13.117***	6.368***	200.572
	(0.167)	(0.237)	(145.197)
San Joaquin	12.475***	6.288***	110.236
	(0.174)	(0.246)	(154.004)
San Luis Obispo	13.943***	5.847***	109.273
	(0.217)	(0.307)	(205.339)
San Mateo	14.021***	6.485***	141.582
	(0.156)	(0.221)	(145.197)
Santa Barbara	13.720***	5.944***	108.140
	(0.170)	(0.242)	(155.222)
Santa Clara	14.171***	5.493***	122.940
	(0.124)	(0.176)	(113.902)
Santa Cruz	14.012***	3.892***	139.707
	(0.276)	(0.391)	(246.407)
Shasta	12.472***	4.536***	139.374
	(0.269)	(0.382)	(197.284)
Siskiyou	11.882***	6.034***	1143.690***
	(0.220)	(0.312)	(185.736)
Solano	13.837***	5.298***	173.224
	(0.243)	(0.345)	(232.833)
Sonoma	13.998***	4.554***	145.786
	(0.152)	(0.216)	(126.404)
Stanislaus	13.695***	6.583***	118.777
	(0.193)	(0.273)	(174.236)
Sutter	13.377***	6.824***	139.874
	(0.365)	(0.517)	(262.671)
Tehama	12.896***	2.542***	163.460
	(0.447)	(0.634)	(308.009)
Tulare	13.075***	5.524***	170.015
	(0.264)	(0.374)	(194.802)
Tuolumne	12.959***	5.716***	167.617
	(0.338)	(0.479)	(308.009)
Yolo	13.262***	5.083***	128.558
	(0.316)	(0.448)	(290.394)
Yuba	12.420***	6.142***	206.696
	(0.381)	(0.540)	(355.658)
N	1,495	1,495	1,806
R^2	0.991	0.913	0.038

M Connection to CMS quality measures

In this section, I describe the CMS quality measures, describe and interpret the results of descriptive regressions, report the regression results, then list references for further information.

M.1 Description and discussion of results

CMS tracks 4 broad quality measures for hospices. I describe them below, using references listed in Appendix M.3:

1. The CAHPS Hospice survey is a list of questions given to the primary caregiver of the hospice patient about the hospice. It asks questions regarding: Communication with Family, Getting Timely Help, Treating Patient with Respect, Emotional and Spiritual Support, Help for Pain and Symptoms, Training Family to Care for Patient, Rating of this Hospice, and Willingness to Recommend this Hospice.

   Unfortunately for my purposes, it began in 2018, and furthermore is missing survey data for the majority of hospices; this is because CMS does not report data if a hospice has not received at least 75 survey responses. For the 12/2019 snapshot, only 262 out of nearly 900 hospices in California have scores reported. Furthermore, the results reported seem to be cumulative data – for instance, the 12/2019 snapshot reflects all survey responses received from 1/2017 to 12/2018. So it is not clear how to parse out across-year quality variation either.

   I merge the 12/2019 snapshot of this survey (containing quality data upto 2018) with my 2018 hospice data including the full list of California counties. I use fuzzy-matching on hospice names. Among the observations that I can match, around 150 have reported scores.

   I run some descriptive regressions, with the results reported below. The results are very noisy and rely on small sample size, so they should be interpreted with caution. I find that visits-per-patient-day is positively correlated with the "ratings" by family caregiver for a wide range of specifications; however it is not statistically significant, although the t-stat is around or greater than 1 in most cases. When I regress log of market share on visits-per-patient-day and ratings, I find that visits-per-patient-day is more strongly related to market share than ratings (based on their t-stats).

   When I merge the survey data with my limited subset of 39 counties, there are few observations remaining (around 50). For that smaller dataset, the above regression results do not

hold, but it is not clear why.

Note also that it is not clear review-based quality measures are necessarily better than input-based quality measures, since they are affected by a different set of issues (such as those detailed in Tadelis (2016)).

2. Hospice Visits in Last Days of Life: Proportion of patients who received a visit in last 2 of 3 days of life. This is very similar to my measure of visits-per-patient-day.

3. "Hospice Care Index": This tracks 10 quality measures – (from the manual) CHC or GIP Provided, Gaps in Skilled Nursing Visits, Early Live Discharges, Late Live Discharges, Burdensome Transitions (Type 1) - Live Discharges from Hospice Followed by Hospitalization and Subsequent Hospice Readmission, Burdensome Transitions (Type 2) - Live Discharges from Hospice Followed by Hospitalization with the Patient Dying in the Hospital, Per-beneficiary Medicare Spending, Skilled Nursing Care Minutes per RHC Day, Skilled Nursing Minutes on Weekend RHC Days, and Visits Near Death.

   This was first published in 2022. This is estimated out of claims data, and so is not subject to the same issues as the CAHPS survey.

   Importantly, several of these measures should correlate with my quality measure of visits-per-patient-day.

   From the manual: "Hospices' HCI scores will depend on their summative performance scores on the component indicators included in the index. Each indicator represents a particular care practice of concern, as identified by CMS/s information gathering and measure testing activities. The indicators are assigned a threshold that determines whether that indicator contributes to the hospice/s total index score. These thresholds reflect either normative performance standards (such as providing certain kinds of services) or reference thresholds relative to national hospice performance (such as performing higher or lower than a certain percentage of all hospices). The thresholds have been set based on CMS's statistical analysis of national hospice performance to ensure validity, reliability, and meaningful distinction between hospices... Hospices' index scores are tied to the number of indicator thresholds they meet. Hospices score better if they meet more thresholds for indicator credit and score worse if they meet fewer indicator thresholds. Each indicator for which a hospice meets or exceeds the "Index Earned Point Criterion" adds 1 point to the hospice's overall HCI score. A hospice's HCI score could range from a perfect 10 (if a hospice received credit for all indicator thresholds) to a 0 (if a hospice failed to meet all performance hresholds)."

   The 2022 document above reports that 85% of the hospices scored 8 or higher out of 10, with 15% hospices scoring lower. It is not clear to me if this HCI is a fine enough measure of

quality because of the threshold/indicator nature of measurement. It certainly helps identify the worst-performing hospices. A big issue with using this quality measure is that it is likely contaminated by COVID restrictions through 2022 and later. More recent data should be free from such problems.

CMS now makes the underlying data for the HCI scores available for the recent few years (all beyond 2018).

4. "HIS Comprehensive Assessment at Admission": This measures the proportion of patients for whom the hospice performed the following 7 care processes – Beliefs/values addressed, Treatment preferences, Pain screening, Pain assessment, Dyspnea treatment, Dyspnea screening, and Treated with an opioid who are given a bowel regimen. To the best of my knowledge, this was first published in 2017 (https://healthcareprovidersolutions.com/new-hospice-comprehensive-assessment-measure-added-to-hospice-compare/) and so is mostly outside my dataset's coverage.

   Importantly, these measures are assessed within the first few days of patient admission, and are not necessarily indicative of the quality of care throughout the patient's stay. So it is unclear how appropriate this is for my setting by itself.

M.2 "Ratings" data and regression results

The CAHPS data reports "ratings" data from caregivers of the hospices' patients. Each hospice is rated as "9 or 10", "7 or 8", and "6 or lower". For each rating tier, the data reports the percentage of caregivers/family members who rated the hospice in that bracket.

For each hospice, I construct a Hospice Compare Score = (Fraction of caregivers choosing highest tier x 10) + (Fraction of caregivers choosing middle tier x 5) + (Fraction of caregivers choosing lowest tier x 1)

The results are summarized and discussed in the previous subsection.

Table 44: Regression of hospice compare score on visits-per-patient-day and other controls, using all counties in California.

	Hospice Compare Score				
	(1)	(2)	(3)	(4)	(5)
(Intercept)	8.600***	8.681***	8.712***	8.770***	8.128***
	(0.219)	(0.173)	(0.154)	(0.139)	(0.472)
Visits-per-patient-day	-0.062	0.181	0.180	0.200	0.145
	(0.200)	(0.167)	(0.161)	(0.161)	(0.200)
For-profit: 1		-0.405***	-0.364***	-0.325***	-0.215*
		(0.049)	(0.067)	(0.084)	(0.090)
Market size			-0.000		
			(0.000)		
log(total firms)				-0.044	-0.029
				(0.023)	(0.029)
Avg. length of stay					0.002
					(0.003)
Hospice inpatient unit: YES					0.312**
					(0.096)
Pediatric program: YES					-0.189*
					(0.088)
Bereavement for non-hospice survivors: YES					0.198***
					(0.045)
Adult day care: YES					0.136
					(0.378)
Agency type: freestanding					0.338
					(0.333)
Agency type: home-health based					0.072
					(0.426)
Agency type: hospital-based					0.429
					(0.357)
Obs.	157	157	157	157	157

Notes: Standard errors clustered at county-level in parentheses.

Table 45: Regression of market share on hospice compare score and visits-per-patient-day, using all counties in California.

	log(market share)	
	(1)	(2)
(Intercept)	-0.198	0.535
	(1.489)	(1.421)
Visits-per-patient-day	0.500	0.451
	(0.287)	(0.338)
Hospice Compare Score	-0.045	-0.140
	(0.145)	(0.128)
For-profit: 1	-0.467**	-0.324*
	(0.146)	(0.152)
Avg. length of stay	-0.011*	-0.012
	(0.005)	(0.006)
log(total firms)	-0.728***	-0.737***
	(0.047)	(0.052)
Hospice inpatient unit: YES		0.369
		(0.206)
Pediatric program: YES		0.336
		(0.308)
Bereavement for non-hospice survivors: YES		-0.057
		(0.104)
Adult day care: YES		0.399
		(0.494)
Agency type: freestanding		0.193
		(0.249)
Agency type: home-health based		-0.507
		(0.395)
Agency type: hospital-based		0.186
		(0.392)
Obs.	157	157

Notes: Standard errors clustered at county-level in parentheses.

M.3 Links to hospice quality measurement

For additional information, please see:

1. Descriptions of CMS hospice quality measures can be found at: https://www.cms.gov/medicare/quality/hospice/current-measures.
2. The archived snapshots of the datasets can be found at: https://data.cms.gov/provider-data/archived-data/hospice-care.
3. https://www.ahrq.gov/talkingquality/measures/setting/long-term-care/hospice.html
4. https://www.cms.gov/files/document/hospice-care-index-hci-technical-reportjuly-2022.pdf.